Algebra II

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1 Category Theory

Suggested papers:

1. Eilenberg and MacLane: *General Theory of Natural Equivalences*
2. B. Mazur: *When is one thing equal to some other thing?*

The first object we study in this course is a *category*. While quite abstract, this object allows a common language to use in mathematics to describe equivalences (e.g. natural transformations vis à vis "unnatural" transformations) and relations between mathematical structures (functors).

Definition 1. A *category* $\mathcal{C}$ consists of a class of objects $\mathrm{ob}(\mathcal{C})$ and for every pair of objects a set[1] $\mathrm{Hom}_{\mathcal{C}}(A, B)$. In the case the pair $(A, B) \neq (C, D)$ these morphisms are disjoint sets. Finally, for every triple $A, B, C \in \mathrm{ob}(\mathcal{C})$ we have a map of sets $\mathrm{Hom}(A, B) \times \mathrm{Hom}(B, C) \to \mathrm{Hom}(A, C)$ satisfying the following two properties:

1. For all $A \in \mathrm{ob}(\mathcal{C})$, there is $\mathrm{id}_A \in \mathrm{Hom}(A, A)$ such that for all $B \in \mathrm{ob}(\mathcal{C})$ and $g \in \mathrm{Hom}(A, B)$, $g = g \circ \mathrm{id}_A$ and for $f \in \mathrm{Hom}(B, A)$, $f = \mathrm{id}_A \circ f$.
2. Composition is associative.

I assume here the reader is familiar with common examples of categories, such as <u>Set</u>, <u>Mon</u> (for monoids), <u>Grp</u>, and <u>Ab</u> (for abelian groups).

N.B. The existence of morphisms between objects in a category is not necessarily guaranteed unless one looks at the morphisms of an object to itself, in which case the identity morphism always exists.

We can express statements about morphisms in terms of commutative diagrams. For example, the identity can be expressed saying the following diagram commutes:

$$\begin{tikzcd}
B \arrow[d, "g"] \arrow[dr, "g"] & \\
A \arrow[r, "\mathrm{id}_A"] \arrow[dr, "f"'] & A \arrow[d, "f"] \\
& B
\end{tikzcd}$$

and the associativity of composition can be expressed as the commutivity of

[1] It is not necessary that this is a set; it could very well be a class. With this restriction, we are technically defining a *locally small* category. When $\mathrm{ob}(\mathcal{C})$ is a proper set, we call $\mathcal{C}$ a *small category*.

$$\begin{tikzcd}
 & B \arrow[r, "g"] \arrow[dr, "h\circ g"] & C \arrow[dr, "h"] & \\
A \arrow[ur, "f"] \arrow[urr, "g\circ f"] \arrow[rrr, "h\circ g\circ f"'] & & & D
\end{tikzcd}$$

1.1 Subcategories

Just as with other mathematical machinary, such as sets, groups, or topological spaces, we should have a way of restricting the focus of a category. We do this by way of *subcategories*.

Definition 2. A category $\mathcal{D}$ is a *subcategory* of $\mathcal{C}$, if $\mathrm{ob}(\mathcal{D})$ is a subclass of $\mathrm{ob}(\mathcal{C})$, the morphisms between objects in $\mathcal{D}$ are morphisms between the same objects in $\mathcal{C}$ (i.e. $\mathrm{Hom}_{\mathcal{D}}(A, B) \subset \mathrm{Hom}_{\mathcal{C}}(A, B)$), and the composition law in $\mathcal{D}$ is the one inherited from $\mathcal{C}$.

N.B. In class it was asked whether subcategories also have to inherit the same identity elements; it was stated that it must, but this does not seem immediate from definition

Thus, we see that subclasses are simply restrictions of a cateogry both in the objects and the morphisms; the compositional structure of the arrows are preserved. In the case the subcategory only restricts the objects and not the morphisms, we say $\mathcal{D}$ is a *full* subcategory of $\mathcal{C}$ (i.e. $\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$ whenever the left-hand side is defined).

N.B. The compositional structure of the morphisms of an object to itself should look familiar: it is a monoid. Moreover, from any monoid M, we can create a category $\underline{M}$ with one object whose morphisms give the monoid M.

1.2 Special Morphisms

A special type of morphism in a category is an *isomorphism*. In the case of cateogries one has previously encountered in math, these are exactly the same as isomorphisms in the traditional sense.

Definition 3. We say $f \in \mathrm{Hom}(A, B)$ is an *isomorphism* if there is $g \in \mathrm{Hom}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

N.B. Note that by the uniqueness of the identity for an object in a category, the above is equivalent to specifying the morphism has a left and right inverse (which *a priori* may not be the same).

Especially in the case of constructing a category from a monoid (and giving no care as to what the object of said category was), one has the feeling that the most important bits of the cateogry are the morphisms. As we shall see later on the course, this is correct. Indeed, by Yonneda, the morphisms in some sense determine the object.

Following this theme of utilizing the importance of morphisms, one can create a new category from an old by merely reversing the direction of arrows in the category. The result of this action is the opposite category (denoted $\mathcal{C}^{op}$). This category has some use when we define contravariant functors.

Definition 4. A morphism f is called *monic* if it satisfies the left-cancellation law—$f \cdot g_1 = f \cdot g_2 \Rightarrow g_1 = g_2$.

A morphism f is called *epic* if it satisfies the right-cancellation law—$g_1 \cdot f = g_2 \cdot f \Rightarrow g_1 = g_2$.

N.B. In the case of some categories (like Set), the notion of monic and epic exactly corresponds to injectivity and surjectivity. However, as we will see below, this does not always hold (as in Ring).

Here are some easy facts that are proven as soon as one writes them down.

Facts.

1. Epic and monic are preserved under composition.
2. If $g \cdot f$ is monic, so is f.
3. If $g \cdot f$ is epic, so is g.
4. If f has a left inverse, it is monic.
5. If f has a right inverse, it is epic.

We now show an example in which monic and epic indeed correspond to injective and surjective:

Proposition 1. *Fix R a ring. In R-mod, monic and epic corresponds to injective and surjective respectively.*

Proof. The fact injective and surjective imply monic and epic respectively follows by passing through the case in Set.

Suppose that $f : A \to B$ is not injective. Then $\ker f$ is an R-mod, and we have morphisms $i : \ker f \to A$ by inclusion and $z : \ker f \to A$ by the 0-map. It is immediate to see these are different morphisms, as $\ker f \neq 0$. However, $f \cdot i = f \cdot z$, implying f is not monic.

Likewise, if $f : A \to B$ is not surjective, the cokernel $B/f(A)$ is non-zero, and the morphisms $B \to B/f(A)$ by the 0-map and projection are not equal, but their compositions with f are; thus, f is not epic. $\square$

N.B. Note that much of the above proof also applies to Grp, except epic implying monic, as $f(A)$ may not be a group. A bit more care is needed to deal with this case, but it still holds.

In the category Ring the above follows except epic may not imply surjective. The proof is the same, expect to show monic implies injective, one uses the subring of $A \times A$ consisting of pairs (a, b) where $f(a) = f(b)$ and the morphisms of projection onto the first and second coordinate, respectively. The classic example of an non-surjective epimorphism is $\mathbb{Z} \hookrightarrow \mathbb{Q}$. In general, when we have a morphism $R \to T$ where the only new data in $T \setminus R$ consists only of inverses of elements of R we have an epimorphism—the structure of the ring and homomorphism determines what these inverse get sent to.

1.3 Functors and Naturality

"The engineer exists!"

Cogdell

Definition 5. A *covariant functor* F between categories $\mathcal{C}, \mathcal{D}$ consists of a map $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ and for every pair of $A, B \in \mathrm{ob}(\mathcal{C})$ a map $F : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{D}}(F(A), F(B))$ satisfying:

1. $F(f \cdot g) = F(f) \cdot F(g)$ whenever $f\cdot$ is defined.
2. $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.

This definition is the same as saying

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f\cdot g=h"'] & B \arrow[d, "f"] \\ & C \end{tikzcd}$$

implies

$$\begin{tikzcd} F(A) \arrow[r, "F(g)"] \arrow[dr, "F(f)\cdot F(g)=F(h)"'] & B \arrow[d, "F(f)"] \\ & F(C) \end{tikzcd}$$

Definition 6. A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ consists of $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ and a map on morphisms $F : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{D}}(F(B), F(A))$ such that

1. $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.
2. $F(f \cdot g) = F(g) \cdot F(f)$.

This composition law can be written as

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f\cdot g=h"'] & B \arrow[d, "f"] \\ & C \end{tikzcd}$$

implies

$$\begin{tikzcd} F(A) & F(B) \arrow[l, "F(g)"'] \\ & F(C) \arrow[u, "F(f)"'] \arrow[ul, "F(h)=F(g)\cdot F(f)"] \end{tikzcd}$$

Example 1. The identity functor $1_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$.

Example 2. If $\mathcal{D}$ is a subcategory of $\mathcal{C}$, we have an *injection functor* $\mathcal{D} \hookrightarrow \mathcal{C}$

Example 3. A class of functors are *forgetfull functors* which basically forget certain structures in the objects of the category.

For an example, $\underline{Mon} \to \underline{Set}$ is given by $F(M)$ is the underlying set of the monoid M. For $f \in \hom_{\underline{Mon}}(A, B)$, $F(f)$ is just viewed as f as a set map.

However, the functor on the morphisms are largely non-surjective.

Similarly we have forgetful functors
$\underline{Grp} \to \underline{Set}$
$\underline{Ring} \to \underline{Set}$
$\underline{Ring} \to \underline{Ab}$ by forgetting the multiplicative structure,
and $\underline{Ring} \to \underline{Mon}$ by forgetting the additive structure.

Example 4. We have $\underline{Grp} \to \underline{Ab}$ by $F(G) = G/(G, G)$ and $f : G \to H$ is sent to $F(f) : G/(G, G) \to H/(H, H)$ by the induced map.

Example 5. We have $\underline{Ring} \to \underline{Grp}$ by $F(R) = R^{\times} = U(R)$.

Here are some examples of *contravariant functors*:

Example 6. The *opposition* or *dual functor* $\mathcal{C} \to \mathcal{C}^{op}$ is defined by $F(A) = A$ and $F(g) = g$.

N.B. One can compose functors to get another functor.

By composing with the opposition functor, any covariant functor $\mathcal{C} \to \mathcal{D}$ can be made into a contravariant functor $\mathcal{C}^{op} \to \mathcal{D}$.

Example 7. Another duality functor is $D : R - \underline{mod} \to \underline{mod} - R$ by $M \mapsto M^* = \hom(M, R)$.

Note that M^* is naturally a right R-module—we define addition and scalar multiplication by $(f + g)(x) = f(x) + g(x)$ and $(f \cdot r)(x) = f(x) \cdot r$.

The functor on the morphisms acts as follows—For $L \in \hom(M, N)$, $D(L) : \hom(N, R) \to \hom(M, R)$ by $D(L)(\phi) = \phi \circ L$. Sometimes we denote $D(L) = L^*$, and it is the *transpose*.

Definition 7. A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if the map on morphisms is injective.

Example 8. All the forgetful functors are faithful—the morphisms are exactly determined by their actions on the underlying sets.

Definition 8. A functor is *full* if the map on morphisms is surjective.

Example 9. The duality functor $D : R - \underline{mod} \to \underline{mod} - R$ is faithful and full.

Definition 9. We say functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are *inverses* if $G \circ F = 1_{\mathcal{C}}$ and $F \circ G = 1_{\mathcal{D}}$.

Definition 10. If $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{C} \to \mathcal{D}$ are covariant functors, a *natural transformation*[2] $\eta : F \to G$ assigns for every $A \in \mathrm{ob}(\mathcal{C})$ a morphism $\eta_A \in \hom_{\mathcal{D}}(F(A), G(A)$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[d, "f"] & F(A) \arrow[r, "\eta_A"] \arrow[d, "F(f)"] & G(A) \arrow[d, "G(f)"] \\ B & F(B) \arrow[r, "\eta_B"] & G(B) \end{tikzcd}$$

This means $G(f) \circ \eta_A = \eta_B \circ F(f)$ for all $f \in \hom_{\mathcal{C}}(A, B)$.

In the case η_A is an isomorphism for all $A \in \mathrm{ob}(\mathcal{C})$, η is called a *natural isomorphism*[3].

In this case, we can define an inverse natural transform $\eta^{-1} : G \to F$ by $(\eta^{-1})_A = \eta_A^{-1}$.

Easily, we can see there is an identity natural transformation for any functor.

We now want to show that there is a natural transformation from the identity functor to the double dual functor.

Example 10. The original example from Eilenberg Mac Lane. We have the contravariant duality functor $D : R - \underline{mod} \to \underline{mod} - R$ by $M \mapsto \hom(M, R)$.

$$\begin{tikzcd} M \arrow[r, "D"] \arrow[d, "L"] & M^* \arrow[r, "D"] & M^{**} \arrow[d, "L^{**}"] \\ N \arrow[r, "D"] & N^* \arrow[r, "D"] \arrow[u, "L^*"] & N^{**} \end{tikzcd}$$

So D^2 is contravariant, $D^2(M) = M^{**}$ Recall the definition of $L^* : g \cdot L = L^*(g)$.

This means the definition of L^{**} is

[2] Also known as a "natty transform"

[3] A.K.A a "natty ice"

$$\begin{tikzcd} N^* \arrow[r, "L^*"] \arrow[dr, "L^{**}(\phi)=\phi\cdot L^*"'] & M^* \arrow[d, "\phi"] \\ & R \end{tikzcd}$$

If $g : N \to R$ (i.e. $g \in N^*$) then $(L^{**}\phi)(g) = \phi(L^*(g)) = \phi(g \cdot L)$.

If $x \in M$ then x definies an element $\eta_M(x) \in M^{**}$ by evaluation—$\eta_M(x)(f) = f(x)$ where $f \in \hom(M, R)$. The claim is that η_M defines a natural transformation $\mathrm{id}_{R-\underline{mod}} \to D^2$.

$$\begin{tikzcd} M \arrow[d, "L"] & M \arrow[r, "\eta_M"] \arrow[d, "L"] & M^{**} = D^2(M) \arrow[d, "L^{**}"] \\ N & N \arrow[r, "\eta_N"] & N^{**} = D^2(N) \end{tikzcd}$$

We have η a natural transform if this diagram commutes for all M, N, L. Let $x \in M$. $L(x) \in N$. Then $\eta_N(L(x))(g) = g(L(x)) = (gL)(x) \in R$. In the other direction if $\phi = \eta_M(x) \in M^{**}$, then

$$L^{**}(\eta_M(x))(g) = \eta_M(x)(L^*g) = (L^*g)(x) = (gL)(x)$$

Therefore the diagram commutes and D^2 is a natural transformation.

N.B. Note that if $R = k$ is a field, then $K - \underline{mod} = K - \underline{vect}$. If moreover we restrict to finite dimensional case, we actually get a natural isomorphism.

Also, the signle dual operation is not natural, as it requires choosing a basis.

Example 11 (Abelianization)**.** $F : \underline{Grp} \to \underline{Ab}$ by $G \mapsto G/(G, G) = G^{ab}$. We can compose this with the injection functor to get $F' : \underline{Grp} \to \underline{Ab} \to \underline{Grp}$. Letting $\nu_G : G \to G^{ab}$ we have the following commuting diagram:

$$\begin{tikzcd} G \arrow[r, "\nu_G"] \arrow[d, "f"] & G^{ab} \arrow[d, "F'(f)"] \\ H \arrow[r, "\nu_H"] & H^{ab} \end{tikzcd}$$

ν then defines a natural transformation.

1.4 Isomorphisms and Equivalences of Categories

Definition 11. We say categories $\mathcal{C}, \mathcal{D}$ are *isomorphic* if we have functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F = 1_{\mathcal{C}}$ and $F \circ G = 1_{\mathcal{D}}$.

This is a really strong equivalence, and when we do so, we identify the categories.

Example 12. $\underline{Ab}$ is isomorphic to $\mathbb{Z} - \underline{mod}$.

$R - \underline{mod}$ is isomorphic to $\underline{mod} - R^{op}$ wher R^{op} just reverses the multiplication operation.

The more interesting notion of equivalence is *equivalence of categories.*

Definition 12. $\mathcal{C}, \mathcal{D}$ are *equivalent* if $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ and $G \circ F \cong 1_{\mathcal{C}}$ and $F \circ G \cong 1_{\mathcal{D}}$ where $\cong$ indicates a natural isomorphism.

N.B.
- Isomorphic categories are equivalent.

- Equivalence of cateogries is an equivalence relation. This really says we can compose natural isomorphisms.

Given an isomorphism of categories, then F and G are honest inverses and F determines G. However, this is not true for a simple equivalence—F can vary by isomorphisms on its map of objects. Can one tell, just on level of F whether F determines an equivalence of categories? The answer is yes, as the following proposition gives us.

Proposition 2. *$F : \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if and only if F is fully faithful and essentially surjective*

Proof. If F is an equivalence of categories, $G \circ F \cong 1_{\mathcal{C}}$ then $\hom_{\mathcal{C}}(A, B) \cong \hom_{\mathcal{C}}(GF(A), GF(B))$ So F must by faithful. Similarly $F \circ G \cong 1_{\mathcal{D}}$ $\hom_{\mathcal{D}}(A, B) \cong \hom_{\mathcal{D}}(FG(A), FG(B))$, so F is surjective on the morphisms and full. Finally, if $A' \in \mathrm{ob}(\mathcal{D})$ then $F \circ G \cong 1_{\mathcal{D}}$, so there is an isomorphism $\eta_{A'}$ in $\mathcal{D}$ between $FG(A')$ and A'. This gives that F is essentially surjective (its range is surjective up to isomorphisms in $\mathcal{D}$.)

We construct $G : \mathcal{D} \to \mathcal{C}$ determining an equivalence of categories with F. SKETCH: Use 3 to define $\mathcal{D}$ on objects—for $A' \in \mathrm{ob}(\mathcal{D})$ take $A \in \mathrm{ob}(\mathcal{C})$ with $F(A)$ isomorphic to A'. Define $G(A') = A$. On morphisms, we have F fully faithful, so we can uniquely find inverses. □

$$\begin{tikzcd} A' \arrow[r, "\eta_{A;}"] \arrow[d, "f'"] & F(A) \arrow[d, dotted, "F(f)"] \\ B' \arrow[r, "\eta_{B'}"] & F(B) \end{tikzcd}$$

$\eta_{B'} \circ f' \circ \eta_{A'}^{-1} \in \hom(F(A), F(B))$. Then use fully faithful to say $\exists! f \in \hom(A, B)$ so that the above diagram commutes. Take $G(f') = f$.

One then must check G defines a functor and (F, G) give equivalences.

Example 13. Let R be a ring, $M_n(R)$ $n \times n$ matrices over R. Then $\underline{mod} - R$ and $\underline{mod} - M_n(R)$ are equivalent:

Proof. $F : \underline{mod} - R \to \underline{mod} - M_n(x)$ by M R-module, let $M^{(n)} = M \oplus \cdots \oplus M$. Say $F(M) = M^{(n)}$ and for $f \in \hom(M, N)$ let $F(f) = f^{(n)}$ where $f^{(n)}(m_1, \ldots, m_n) = (f(m_1), \ldots, f(m_n))$. Check this is fully faithful and essentially surjective. □

"I put it on Fridays just because I do" "The answer is no because we have class on that Friday and that would be evil"

1.5 Products and Co-products

> I don't want to say you can't do that,
> but it hurts my head.
>
> *Cogdell*

The main idea is to generalize the idea of the cartesian product of sets or groups to arbitrary sets. However, the classical product is defined in terms of pairs of elements, which we don't have in general cateogry theory. Thus we have to determine a way to do this just using morphisms. Once we do this, dualizing gives the coproduct.

Two important properties of the product: If $G = G_1 \times G_2$ there are 2 projections: $G_1 \times G_2 \to G_1$ and $G_1 \times G_2 \to G_2$ and these are morphisms in the category. If H is a group, and morphisms $H \to G_1$ and $H \to G_2$ then there is a unique $f : H \to G_1 \times G_2$ making the morphisms commute with the projections. This is expressed in the following commutative diagram.

$$\begin{tikzcd}
 & H \arrow[dl, "f_1"'] \arrow[d, dotted, "\exists! f"] \arrow[dr, "f_2"] & \\
G_1 & G \arrow[l, "p_1"] \arrow[r, "p_2"'] & G_2
\end{tikzcd}$$

In groups, f must be defined by $f(h) = (f_1(h), f_2(h))$.

With this notion in hand, we define a general product in a category:

Definition 13. Let $A_1, A_2 \in \mathrm{ob}(\mathcal{C})$. The *product* of A_1, A_2, denoted $A_1 \times A_2$ is a triple (A, p_1, p_2) where $A \in \mathrm{ob}(\mathcal{C})$ and $p_i \in \hom_{\mathcal{C}}(A, A_i)$ satisfying the following universal property: If $B \in \mathrm{ob}(\mathcal{C})$ and $f_i \in \hom_{\mathcal{C}}(B, A_i)$, then there is a unique $f \in \hom(B, A_1 \times A_2)$ such that the following diagram commutes:

$$\begin{tikzcd}
 & B \arrow[dl, "f_1"'] \arrow[d, dotted, "\exists! f"] \arrow[dr, "f_2"] & \\
A_1 & A_1 \times A_2 \arrow[l, "p_1"] \arrow[r, "p_2"'] & A_2
\end{tikzcd}$$

The product $A_1 \times A_2$ can be written as A, $A_1 \times A_2$ or $A_1 \prod A_2$.

N.B. In the definition we used the words "universal mapping property"—this corresponds to the existence of a unique morphism between objects in the category. In particular, this gives products in a category is unique up to isomorphisms (if they exist). A big question that remains then is if products exist in a certain category.

Note this definition can also generalize to arbitrary products (not just finite) in the obvious way.

Definition 14. Let $\{A_\alpha\}_{\alpha \in A}$ be an index set of objects of $\mathcal{C}$. A *product* of $\{A_\alpha\}$ is a tuple consisting of the data $A \in \mathrm{ob}(\mathcal{C})$ and morphisms $p_\alpha : A \to A_\alpha$ satisfying the universal property that if $B \in \mathrm{ob}(\mathcal{C})$ with morphisms $f_\alpha : B \to A_\alpha$ there is a unique morphisms $B \to A$ such that $f_\alpha = p_\alpha \circ f$ for all α.

This is a true extension of the definition, as when $|A| = 2$, this is exactly the product defined above. We write $A = \prod_\alpha A_\alpha$, and if it exists it is unique up to isomorphism.

Example 14.

1. Arbitrary products exist in <u>Sets</u>— Given $\{A_\alpha\}_{\alpha \in I}$. Define $A := \left\{a : I \to \bigcup_{\alpha \in I} A_\alpha \text{ so that } a(\alpha) \in A_\alpha\right\}$. Define $p_\alpha(a) = a(\alpha)$.

   We claim this is a product in <u>Set</u>. Given $(B, \{f_\alpha\})$ where $f_\alpha : B \to A_\alpha \in \hom(B, A_\alpha)$. Then we must have $f : B \to A$ by $f(b) \mapsto a$ where $a(\alpha) = f_\alpha(b)$. This clearly makes everything commute, and it must be this, so it is the unique such morphism $B \to A$.

   A can also be realized as tuples of elements of A_α, and this is the way we have typically viewed Cartesian products

2. In <u>Grp</u>, we form $G = \prod_\alpha G_\alpha$ as defined in algebra class (component-wise multiplication).

3. In <u>Ring</u>, the same idea of the product holds.

The idea of the coproduct is the exact dual of the product.

Definition 15. In a category $\mathcal{C}$, the *coproduct* of $\{A_\alpha\} \in \mathrm{ob}(\mathcal{C})$ consists of the data $A \in \mathrm{ob}(\mathcal{C})$ and morphisms $i_\alpha : A_\alpha \to A$ following the universal property— if $B \in \mathrm{ob}(\mathcal{C})$ with morphisms $f_\alpha : A_\alpha \to B$, there is a unique morphism $f : A \to B$ so that $f_\alpha = f \cdot i_\alpha$ for all α.

In this case, we write $A = \coprod_\alpha A_\alpha$. Note by the universal mapping property, if such exists, it is unique up to isomorphism.

The question of existence still exists. Here are some examples for where it holds.

Example 15.

1. In $\underline{\mathrm{Set}}$, coproducts exist and is just the disjoint union and the morphisms are simply the natural injections. This is why our notation of both the coproduct and disjoint union as $\coprod A_\alpha$ makes sense.

2. In $R-\underline{mod}$ we also have the existence of coproducts, but the construction is more complicated. Say $\{M_\alpha\}$ is an indexed set over I of R-modules. Let $M = \prod M_\alpha$ be the product. Set $\bigoplus M_\alpha = \{m : I \to \prod M_\alpha : m(\alpha) = 0 \text{ for all but finite } \alpha\}$. We again have the natural inclusion $i_\alpha : M_\alpha \to \bigoplus M_\alpha$. We claim this is the coproduct.

   Suppose $A \in \mathrm{ob}(R - \underline{mod})$ with morphisms $f_\alpha : M_\alpha \to A$. We define $f : \bigoplus M_\alpha \to A$ by defining $f(0, 0, \dots, m_\alpha, 0, 0, \dots)$ and then expanding linearlly because we are only dealing with finite sums. In fact, this is the only way for which $f \cdot i_\alpha = f_\alpha$. This shows $\bigoplus M_\alpha = \coprod M_\alpha$

   This example is particularly useful once we get to homological algebra. The similar construction also works for abelian groups (as they are $\mathbb{Z}$ modules) as well as arbitrary groups and rings.

1.6 Direct and Inverse Limits

> Why don't they call contravariant functors 'funky functors'?
>
> *Matt*

Let I be a directed set—a set I with a binary relation $\leq$ such that if $\alpha, \beta \in I$

1. $\alpha \leq \alpha$
2. $\alpha \leq \beta$ and $\beta \leq \gamma$ implies $\alpha \leq \gamma$.
3. Given $\alpha, \beta \in I$ there is $\gamma \in I$ such that $\alpha \leq \gamma$ and $\beta \leq \gamma$.

This defines a category $\underline{I}$ where $\mathrm{ob}(\underline{I}) = I$ and $\hom(A, B) = \begin{cases} \emptyset & \alpha \not\leq \beta \\ \{f_{\alpha,\beta}\} & \alpha \leq \beta \end{cases}$.

If $\mathcal{C}$ is any category, consider a convariant functor $F : \underline{I} \to \mathcal{C}$. Then F consists of a map $I \to \mathrm{ob}(\mathcal{C})$ and a map $(\alpha, \beta) \to \phi_{\alpha\beta} = F(f_{\alpha\beta})$ if $\alpha \leq \beta$.

So we have maps

$$A_\alpha \xrightarrow{\phi_{\alpha\beta}} A_\beta$$

when $\alpha \leq \beta$ such that $\phi_{\alpha\alpha} = \mathrm{id}_{F(\alpha)}$ and $\phi_{\beta\gamma} \cdot \phi_{\alpha\beta} = \phi_{\alpha\gamma}$ whenever $\alpha \leq \beta \leq \gamma$.

The "image" of F is called a *directed system* in $\mathcal{C}$ consisting of the data $\{A_\alpha, \phi_{\alpha\beta} : \alpha \leq \beta\}$. This allows us the view objects of $\mathcal{C}$ as living in a sort of lattic or structure mimicking the order of the directed set.

What happens then to the directed system as we move along the path of the system?

Definition 16. Let I be a directed set, $\mathcal{C}$ a category, $F : \underline{I} \to \mathcal{C}$ a covariant functor. Call $F(\alpha) = A_\alpha$ and $\phi_{\alpha\beta} = F(f_{\alpha\beta}) \in \hom(A_\alpha, A_\beta)$ for $\alpha \leq \beta$.

Then a *direct limit* $\lim_{\to_\alpha} A_\alpha = \lim_{\to_\alpha}(A_\alpha, \phi_{\alpha\beta})$ is a set $(A, \eta_\alpha)_{\alpha \in I}$ such that fo $A \in \mathrm{ob}(\mathcal{C})$ $\eta_\alpha \in \hom(A_\alpha, A)$ such that $\eta_\alpha = \eta_\beta \cdot \phi_{\alpha\beta}$.

$$\begin{tikzcd}
A_\alpha \arrow[rr, "\phi_{\alpha\beta}"] \arrow[dr, "\eta_\alpha"'] & & A_\beta \arrow[dl, "\eta_\beta"] \\
& A &
\end{tikzcd}$$

And we require A and $\{\eta_\alpha\}$ is universal in this construction—if $B \in \mathrm{ob}(\mathcal{C})$ and we have $\zeta_\alpha \in \hom(B, A_\alpha)$ satisfying the above commutative diagram, then there is a unique morphism $\theta : A \to B$ such that the following diagram commutes:

$$\begin{tikzcd}
A_\alpha \arrow[rr, "\phi_{\alpha\beta}"] \arrow[dr, "\eta_\alpha"] \arrow[ddr, "\zeta_\alpha"'] & & A_\beta \arrow[dl, "\eta_\beta"'] \arrow[ddl, "\zeta_\beta"] \\
& A \arrow[d, "\theta"] & \\
& B &
\end{tikzcd}$$

Therefore if directed limits exists, they are unique (up to isomorphism). Directed limits are also called *inductive limits*.

Here are some examples of when these direct limits exist:

Example 16. Let $\mathcal{C} = \underline{Set}$ and $I = \mathbb{N}$ with the usual ordering. The directed system take $i \in \mathbb{N}$ a set A_i so that the maps $A_i \hookrightarrow A_{i+k}$ is an injection. Thus we can view $A_i \subset A_{i+k}$

The directed limit is then the union $U_i A_i$ and the morphisms are just inclusions.

Example 17. In $R - \underline{mod}$, begin with a directed system of modules $\{M_\alpha, \phi_{\alpha\beta}\}$. Take $\widetilde{M} = \bigoplus_\alpha M_\alpha = \coprod_\alpha M_\alpha$. Let $N \subset \widetilde{M}$ be the submodule generated by sequences
$(0, 0, \cdots, m_\alpha, 0, \cdots, 0, -\phi_{\alpha\beta}(m_\alpha), 0, \cdots)$ where m_α appears in the α-th position and $-\phi_\alpha\beta(m_\alpha)$ in the β-th position.

Take $M = \widetilde{M}/N = \lim_{to_\alpha} M_\alpha$ where $\eta_\alpha \hookrightarrow \bigoplus_\alpha M_\alpha \to^p \widetilde{M}/N$.

This basically works by identifying objects that get mapped to each other.

Example 18. $\mathcal{C} = \underline{Grp}$ if we have a directed system $\{G_\alpha, \phi_{\alpha\beta}\}$ form $\coprod_\alpha G_\alpha$ in set. Put on an equivalence relation by $g_\alpha \in G_\alpha$ and $g_\beta \in G_\beta$ say $g_\alpha \sim g_\beta$ if there is $\alpha, \beta \leq \gamma$ so that $\phi_{\alpha\gamma}(g_\alpha) = \phi_{\beta\gamma}(g_\beta)$. Thus things are equivalent if, going far enough out, elements are mapped to the same thing.

Let $G = (\coprod G_\alpha)/\sim$ and $\eta_\alpha : G_\alpha \to G$ by $g_\alpha \mapsto [g_\alpha]$. Check the obvious group structure is well-defined and gives us a direct limit.

This same structure also works for $\underline{Ring}$ and such.

Definition 17. I directed set, $F : \underline{I} \to \mathcal{C}$ a contravariant functor. The "image" of F is called an *inverse system*. An *inverse limit* of an inverse system is an object $A \in \mathrm{ob}(\mathcal{C})$ along with morphisms $\eta_\alpha : A \to A_\alpha$ making the following diagram commute

$$\begin{tikzcd} A_\alpha \arrow[rr, "\phi_{\alpha\beta}"] & & A_\beta \\ & A \arrow[ul, "\eta_\alpha"] \arrow[ur, "\eta_\beta"'] & \end{tikzcd}$$

and satisfying the universal property that if B is another objects with morphisms $\zeta_\alpha : B \to A_\alpha$ making the above diagram commute then there is a morphism $\theta : B \to A$ so that $\zeta_\alpha = \eta_\alpha \circ \theta$.

We denote the inverse limit as $\lim_{\leftarrow_\alpha} A_\alpha$ or $\lim_{\leftarrow_\alpha} \{A_\alpha, \phi_{\beta\alpha}\}$.

Again, once we have existence, uniqueness follows from the universal mapping property. Inverse limits are also know as *projective limits*

An important example of this sort of limit is the *p-adic* integers.

Example 19. What do inverse limits look like in $\underline{\text{Grp}}$? It looks like Γ which consists of sequences in $\prod G_\alpha$ so that the elements of the sequence are compatible with the homomorphisms, i.e., $\phi_{\alpha\beta}(g_\alpha) = g_\beta$ for all $\alpha \geq \beta$. The maps $\eta_\alpha : G \to G_\alpha$ is projection on the α-th position.

Note then that the sequence of identities is in this group. Component-wise multiplication makes this into a group, as $\phi_{\alpha\beta}$ are homomorphisms.

Example 20. Fix G a group. Let $I = \{H \triangleleft G : (G : H) < \infty\}$. Order I by reverse inclusion (this works as $H \cap K \triangleleft G$ and is of finite index). Maps $\phi_{KH} : G/K \to G/H$ is then an inverse system in $\underline{\text{Grp}}$.

The inverse limit $\lim_{\leftarrow_H} G/H$ then exists, and the resulting group is called the *completion* of G. This is what is known as a *profinite group*—an inverse limit of finite groups.

We can embed $G \hookrightarrow \lim_{\leftarrow_H} G/H$ if $\bigcap_H H = \{e\}$.

1.7 The Hom Functors, Representable Functors, and Yoneda Lemma

> "That's why you should get a Mastercard"
>
> James in response to "One student is still having visa problems"

Given a category $\mathcal{C}$, there are two types of natural functors $\mathcal{C} \to \underline{Set}$. The first is covariant: $\hom(A, -)$ for fixed $A \in \text{ob}(\mathcal{C})$. The second is contravariant $\hom(-, A)$.

N.B. We can also view this is a bi-functor $\hom_{\mathcal{C}}(-,-) : \mathcal{C}^{op} \times \mathcal{C} \to \underline{Set}$.

Definition 18. A covariant functor $F : \mathcal{C} \to \underline{Set}$ is called *representable* if it is representable as a hom functor, i.e., $F \cong \hom(A, -)$ for some $A \in \text{ob}(\mathcal{C})$.

Similarly, a contravariant functor $G : \mathcal{C} \to \underline{Set}$ is *representable* if naturally isomorphic to $\hom(-, A)$ for some $A \in \text{ob}(\mathcal{C})$.

In either case, we call A a *representative* of the functor.

This definition leads to the following question: In what cases do we have a natural transformation $\eta : \hom(A, -) \to F$ for a covariant functor $F : \mathcal{C} \to \underline{Set}$?

If such η exsits, we always have a set map $\eta_B : \hom(A, B) \to F(B)$ such that for all $g \in \hom(B, C)$ we have the commuting diagram

$$\begin{tikzcd} \hom(A,B) \arrow[r, "\eta_B"] \arrow[d, "\hom(A,g)"] & F(B) \arrow[d, "F(g)"] \\ \hom(A,C) \arrow[r, "\eta_C"] & F(C) \end{tikzcd}$$

We also have an evaluation map for any $a \in A$ and $B \in \mathrm{ob}(\mathcal{C})$ by $a_B : \hom(A,B) \to F(B)$ by $f \mapsto F(f)(a)$. Thus for this natural transformation to exists, we must have commutivity of the diagram satisfied by the evaluation maps. If $\eta_B = a_B$ then commutativity means for all $f \in \hom(A,B)$ and $g \in \hom(B,C)$, $F(gf)(a) = F(g)(F(f)(a)) = F(g)(a_B(f)) = a_C \hom(A,g)(f) = F(gf)(a)$. This shows $\eta(a) : B \to a_B$ is a natural transformation from $\hom(A,-) \to F$.

N.B. $\eta(a)_A(\mathrm{id}_A) = a$.

It turns out these are all the natural transformations we can get from $\hom(A,-)$ to F.

Lemma 1. *If $\eta : \hom(A,-) \to F$ is a natural transformation, then $\eta = \eta(a)$ for some $a \in F(A)$.*

Proof. Consider $f \in \hom(A,B)$ and the diagram

$$\begin{tikzcd} \hom(A,A) \arrow[r, "\eta_A"] \arrow[d, "\hom(A,f)"] & F(A) \arrow[d, "F(f)"] \\ \hom(A,B) \arrow[r, "\eta_B"] & F(B) \end{tikzcd}$$

Then

$$\eta_B(f) = \eta_B(f \cdot \mathrm{id}_A) = \eta_B(\hom(A,f)(\mathrm{id}_A) = F(f) \cdot \eta_A(\mathrm{id}_A)$$

Take $a = \eta_A(\mathrm{id}_A)$. □

Combining this lemma with the previous discussion, we arrive at Yoneda's Lemma:

Lemma 2 (Yoneda's Lemma)**.** *Let $F : \mathcal{C} \to \underline{Set}$ be a covariant functor and $A \in \mathrm{ob}(\mathcal{C})$. Then the natural transformations of $\hom(A,-)$ to F are in bijection with elements in $F(A)$ by $a \in F(A) \mapsto \eta$ where $\eta_B = a_B$. The inverse is given my $\eta \mapsto \eta_A(\mathrm{id}_A)$.*

If F is representable, with representative A, the natural isomorphism is given by $\eta(a)$ for some $a \in F(A)$. So we can call (A,a) a *representative* of F.

1.8 The Category of Modules

> "I've done almost none of the analysis." "That's a good move."
>
> Cogdell in response to Daniel

Let R be a ring (with identity). We have seen $R - \underline{mod} \cong \underline{mod} - R^{op}$. For this reason we will focus only on left R-modules. This category has a 0-object which is the trivial module. For $M, N \in \mathrm{ob}(\mathcal{C})$ we define the 0-*morphism* to be the composition of $M \to 0 \to N$. In this context, the map sends everything in M to 0_N.

Importantly, each of the morphism sets $\hom_R(M,N)$ carry the addition structure of an abelian group. For $f, g \in \hom_R(M,N)$ we define $f + g \in \hom_R(M,N)$ by $(f+g)(m) = f(m) + g(m)$. The identity in the group is given by the 0-morphism. Thus $(\hom_R(M,N), +, 0)$ is an abelian

group. Moreover, the addition commutes with composition (i.e. $f \cdot (g+h) = f \cdot g + f \cdot h$ and $(f+g) \cdot h = f \cdot h + g \cdot h$ whenever these compostions make sense).

When looking at functors between module categories, we would also like to consider when the functor preserves the group structures on the morphism sets.

Definition 19. A functor $F : R - \underline{mod} \to S - \underline{mod}$ is *additive* if $F(f+g) = F(f) + F(g)$ for $f, g \in \hom_R(M, N)$.

This can be said of both covariant and contravariant functors.

From this point forward, all functors between module categories are assumed to be additive.

N.B. In the case $M = N = P$, then $End(M) := \hom_R(M, M)$ gives a ring structure where multiplication is defined by composition.

Recall that in $R-\underline{mod}$ we have both product and coproducts (because these are abelian groups). In the case the index set I of the (co)product is finite, the product and coproduct over I is the same group, and the injections $i_j : M_j \to M$ and projections $p_i : M \to M_i$. Then $p_j \circ i_j = \mathrm{id}_{M_j}$, $p_k \circ i_j : M_j \to M_k = 0$, and $\sum_{i=1}^n i_i p_i = \mathrm{id}_M$.

Moreover, these properties of morphisms completely determine M. As if we have a module M' with injections $i'_j \in \hom(M_i, M')$ and projections $p'_j \in \hom(M', M_j)$ satisfying the above three properties, then $\theta = \sum_{j=1}^n i'_j p_j \in \hom(M, M')$ and $\theta' = \sum_{j=1}^n i_j p'_j \in \hom(M', M)$. Then $\theta' \cdot \theta \in \hom(M, M)$ is the identity on M and $\theta \cdot \theta'$ is the identity on M'. Therefore the morphisms $\{i_j\}, \{p_j\}$ completely characterize the product and coproduct.

A consequence of this fact is that an additive functor preserves products and coproducts. We can express this as $F(\oplus_{i=1}^n M_i) = \oplus_{i=1}^n F(M_i)$ with projections $F(p_j)$ and injections $F(i_j)$ (in the case of a contravariant functor, the projections are sent to the injections and vice versa).

From these properties of modules, we can define an *additive category*

Definition 20. A category $\mathcal{C}$ is called an *additive category* if

1. $\mathcal{C}$ has a 0-object.
2. For $M, N \in \mathrm{ob}(\mathcal{C})$ we have $\hom_{\mathcal{C}}(M, N)$ has the structure of abelian groups with the identity being the zero morphism.
3. The additive structure distributes over composition ($f \cdot (g+h) = f \cdot g + f \cdot h$ and $(g+h) \cdot f = g \cdot f + h \cdot f$ whenever these compositions are defined.)
4. Given a finite set of objects $A_1, \ldots, A_n \in \mathrm{ob}(\mathcal{C})$ there exists $A \in \mathrm{ob}(\mathcal{C})$ and morphisms $p_j \in \hom_{\mathcal{C}}(A, A_j)$ and $i_j \in \hom_{\mathcal{C}}(A_j, A)$ such that $p_j \cdot i_j = \mathrm{id}_{A_j}$, $p_j \cdot i_k = 0$ for $j \neq k$ and $\sum_{j=1}^N i_j \cdot p_j = \mathrm{id}_A$

1.8.1 Exactness

If M, N, P are R-modules and we have maps

$$M \xrightarrow{f} N \xrightarrow{g} P$$

Is *exact* at N if $\mathrm{im}(f) = \ker(g)$.

The definition extends to longer sequences

$$\cdot \longrightarrow M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \xrightarrow{f_3} M_4 \xrightarrow{f_4} \cdots$$

is *exact* at M_i if $\mathrm{im}(f_{i-1}) = \ker(f_i)$.

Proposition 3. *1. $0 \to M \xrightarrow{f} N$ is exact at M if and only if $\ker(f) = 0$ if and only if f is injective if and only if f is monic.*

2. $M \xrightarrow{g} N \to 0$ is exact at N iff $\mathrm{im}(g) = N$ iff f is surjective iff f is epic.

3. The map $0 \to \ker(f) \to M \xrightarrow{f} N$ is always exact.

4. If we have $M \xrightarrow{f} N$, define $\mathrm{coker}(f) := N/\mathrm{im}(f)$. Then $M \xrightarrow{f} N \to \mathrm{coker}(f) \to 0$ is always exact. This gives f surjective iff $\mathrm{coker}(f) = 0$.

5. If $f : M \to N$ is any morphism in $R - \underline{mod}$, we can always factor $M \to \mathrm{im}(f) \to N$ where the first map is epic and the second is monic.

6. $0 \to \ker(f) \to M \xrightarrow{f} N \to \mathrm{coker}(f) \to 0$ is exact.

Definition 21. An additive category $\mathcal{C}$ is an *abelian* category if

1. Every morphism in $\mathcal{C}$ has a kernel and cokernel.
2. Every monic morphism is a kernel of its cokernel.
3. Every epic morphism is a cokernel of its kernel.
4. Every morphism factors as the composition of an epic followed by a monic.

2 Module Categories

2.1 Exactness and Functors

Let $F : R - \underline{mod} \to S - \underline{mod}$ be an additive functor. We say F is *exact* if is preserves short exact sequences. This means if

$$0 \to M' \to M \to M'' \to 0$$

is exact then so is

$$0 \to F(M') \to F(M) \to F(M'') \to 0$$

(in the case of a contravariant functor the last row of arrows are reversed).

Often, it is enough to consider when a funtor is partially exact, i.e., exact on one side.

If F is covariant, then F is *right exact* if

$$M' \to M \to M'' \to 0$$

exact implies

$$F(M') \to F(M) \to F(M'') \to 0$$

is exact. Similarly, F is *left exact* if

$$0 \to M' \to M \to M''$$

exact implies

$$0 \to F(M') \to F(M) \to F(M'')$$

is exact.

In the case F is contravariant, F is *right exact* if

$$0 \to M' \to M \to M''$$

implies

$$F(M'') \to F(M) \to F(M') \to 0$$

is exact. Similarly, F is called *left exact* if

$$M' \to M \to M'' \to 0$$

implies

$$0 \to F(M'') \to F(M) \to F(M')$$

is exact.

For module cateogries, we have two natural functors

$$\hom(M, -) : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$$

and

$$\hom(-, M) : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$$

Theorem 1. *These functors are left exact.*

Proof. We will prove this for $\hom(M, -)$, and the proof for the contravariant Hom functor is left for homework.

Suppose

$$0 \to N' \xrightarrow{f} N \xrightarrow{g} N''$$

is an exact sequence in $R - \underline{mod}$. Thus f is injective and $\mathrm{im}(f) = \ker(g)$. We first show $\hom(M, f)$ is injective. Indeed, if $h, h' \in \hom(M, N')$ with $f \circ h = f \circ h'$, we have $h = h'$, as f is injective.

Now suppose $h \in \hom(M, N) \in \ker(\hom(M, g))$. Thus, $\mathrm{im}(h) \subseteq \ker(g)$ which implies $\mathrm{im}(h) \subset \mathrm{im}(f)$. As f is injective, for every $m \in M$, we can find a unique $n' \in N'$ so that $h(m) = f(n')$. Thus define a function $h' : M \to N'$ by $m \mapsto n'$. It is easy to check this is a module homomorphism and that $f \circ h' = h$. Thus $h \in \mathrm{im}(\hom(M, f))$.

Now suppose $h \in \mathrm{im}(\hom(M, f))$. thus $h = f \circ h'$ for some morphism $h' : M \to N'$. Then $g \circ h = g \circ f \circ h' = 0 \circ h' = 0$. Thus $\mathrm{im}(\hom(M, f)) \subseteq \ker(\hom(M, g))$. The reverse containments give $\ker(\hom(M, g)) = \mathrm{im}(\hom(M, f))$.

This shows the Hom functor is left exact. □

In the homework, we will also see examples showing these Hom functors are strictly left exact and not exact.

N.B. In the case R is commutative, the $\hom_R(M, N)$ sets also have the structure of an R-module given by $(r \cdot f)(m) = r \cdot f(m)$. Thus we can view $\hom_R(M, -)$ and $\hom_R(-, M)$ as functors to $R - \underline{mod}$ and not just $\mathbb{Z} - \underline{mod}$.

The next example of module functors we consider are the tensor functors $\otimes$—we will see these are right exact.

Definition 22. Let M be a right R-module and N be a left R-module. Define a *Balanced Product* of M, N to be an abelian group P with a map $f : M \times N \to P$ such that

1. $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$
2. $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$
3. $f(mr, n) = f(m, rn)$

A morphism between balanced products of M and N consists of a morphism of abelian groups $\eta : (P, f) \to (Q, g)$ such that $\eta(f(m, n)) = g(m, n)$. Thus the following diagram commutes

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "g"] & P \arrow[dl, "\eta"] \\ Q & \end{tikzcd}$$

A *tensor product* $(M \otimes_R N, \otimes)$ is a balanced product with the following universal mapping property: If (P, f) is any other balanced product of M, N then there is a unique map $\eta : M \otimes_R N \to P$ such that the following diagram commutes

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[d, "f"] & M \otimes_R N \arrow[dl, "\eta"] \\ P & \end{tikzcd}$$

We note that $M \otimes N$ exists, and the universal mapping property gives it is unique up to isomorphism. Also, $M \otimes N = \langle m \otimes n : m \in M \;\; n \in N \rangle$. Elements of the form $m \otimes n$ are caled *pure tensors* or *decomposable elements*.

2.2 Tensor Products

> "I believe you beat that horse with Gaith"
>
> Cogdell

Recall the construction of the tensor product: First, form the free $\mathbb{Z}$-module on $M \times N$. This is just $F = \bigoplus_{M \times N} \mathbb{Z}$. Elements of F take the form $\sum x_i(m_i, n_i)$ where $m_i \in M$, $n_i \in N$ and $x_i \in \mathbb{Z}$. Let A be the submodule of F generated by the relations defining the balanced product: $(m_1 + m_2, n) - (m_1, n) - (m_2, n)$, $(m, n_1 + n_2) - (m, n_1) - (m, n_2)$, and $(mr, n) - (m, rn)$. Then $M \otimes_R N = F/A$ and $m \otimes n = (m, n) + A$.

2.2.1 $\otimes$ as a functor

We have $\otimes : \underline{mod} - R \times R - \underline{mod} \to \mathbb{Z} - \underline{mod}$. If $M \xrightarrow{f} M'$ and $N \xrightarrow{g} N'$ then we have a map

$$M \times N \xrightarrow{f \times g} M' \times N' \xrightarrow{\otimes} M' \otimes N'$$

The composition gives a balanced product on $M \times N$, so it induces a morphism

$$M \otimes N \xrightarrow{f \otimes g} M' \otimes N'$$

On pure tensors, $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$. In the case we have multiple morphisms

$$M \xrightarrow{f} M' \xrightarrow{f'} M''$$

$$N \xrightarrow{g} N' \xrightarrow{g'} N''$$

Then on the generators, we have

$$(f' \cdot f)(\otimes(g' \cdot g)(m \otimes n) = (f'f)(m) \otimes (g'g)(n) = (f' \otimes g') \cdot (f \otimes g)(m\otimes)$$

Whence

$$(f' \cdot f) \otimes (g' \cdot g) = (f' \otimes g') \cdot (f \otimes g)$$

Theorem 2. *The maps $(M \times N) \to M \otimes N$ and $(f \times g) \mapsto f \otimes g$ defines a covariant bifunctor $\underline{mod} - R \times R - \underline{mod} \to \mathbb{Z} - \underline{mod}$.*

By fixing either M or N we get a tensor functor $M \otimes - : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ or a tensor functor $- \otimes N : \underline{mod} - R \to \mathbb{Z} - \underline{mod}$ respectively.

We should also check these functors are additive: Indeed, on generators we have $(f_1 + f_2) \otimes \mathrm{id}_N(m \otimes n) = (f_1 + f_2)(m) \otimes n = f_1(m) \otimes n + f_2(m) \otimes n$ which shows $(M \otimes -)(f_1 + f_2) = (M \otimes -)(f_1) + (M \otimes -)(f_2)$. A similar proof demonstrates the same for $(- \otimes N)$.

This means the $\otimes$ functors preserve finite products and coproducts. However, there is an issue for arbitrary products. This problem does not arise for arbitrary coproducts.

Proposition 4. *If $\{M_\alpha\}_{\alpha \in I}$ a set of right R-modules, and N a fixed left R-modules, then $(\oplus_{\alpha \in I} M_\alpha) \otimes N \cong \oplus_\alpha (M_\alpha \otimes N)$ by $((x_\alpha)_{\alpha \in I} \otimes n) \mapsto (x_\alpha \otimes n)_{\alpha \in I}$.*

Proof. We claim that the defined map is a balanced product, so we have an induced map

$$(\oplus M_\alpha) \otimes N \xrightarrow{\eta_N} \oplus_\alpha (M_\alpha \otimes N)$$

In the other direction, we have the injections $i_\alpha : M_\alpha \to \oplus_\alpha M_\alpha$. Tensoring with the identity gives

$$i_\alpha \otimes \mathrm{id}_N : M_\alpha \otimes N \to (\oplus_\alpha M_\alpha) \otimes M$$

Using the universal mapping property of the coproduct gives a map

$$\oplus_\alpha (M_\alpha \otimes N) \to (\oplus_\alpha M_\alpha) \otimes N$$

One easily checks this is the inverse of η_N (by checking on generators). $\square$

2.2.2 Exactness of $\otimes$ functors

Theorem 3. *The one-sided $\otimes$ functors are both right exact, covariant functors.*

Proof. Consider $M \otimes - : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$. Suppose

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \to 0$$

is an exact sequence.

Consider the exactness of

$$M \otimes N' \xrightarrow{\mathrm{id}_M \otimes f} M \otimes N \xrightarrow{\mathrm{id}_M \otimes g} M \otimes N'' \to 0$$

Suppose $m \otimes n''$ is a pure tensor of $M \otimes N''$. As g is surjective, there is $n \in N$ such that $g(n) = n''$. Thus $\mathrm{id}_M \otimes g(m \otimes n) = m \otimes n''$. This shows $\mathrm{id}_M \otimes g$ is surjective.

Now suppose $\sum m_i \otimes f(n_i')$ is in the image of $\mathrm{id}_M \otimes f$. We can calculate that $(\mathrm{id}_M \otimes g)(\sum m_i \otimes f(n_i')) = \sum m_i \otimes gf(n_i') = \sum m_i \otimes 0 = 0$. Therefore $\mathrm{im}(\mathrm{id}_M \ \ otimesf) \subset \ker(\mathrm{id}_M \otimes g)$.

As $\mathrm{id}_M \otimes g$ is surjective, $M \otimes N'' \cong M \otimes N / \ker(\mathrm{id}_M \otimes g)$ and as $\mathrm{im}(\mathrm{id}_M \otimes f) \subseteq \ker(\mathrm{id}_M \otimes g)$ we can map

$$M \otimes N / \mathrm{im}(\mathrm{id}_M \otimes f) \xrightarrow{\theta} M \otimes N'' \to 0$$

In this case, the statement $\mathrm{im}(\mathrm{id}_M \otimes f) = \ker(\mathrm{id}_M \otimes g)$ is the same as θ being an isomorphism. Therefore, we wil show θ is an isomorphism, by constructing θ' its inverse.

For $M \times N''$, we can define $\theta'(m, n'') = m \otimes n + \mathrm{im}(\mathrm{id}_M \otimes f)$ where $n \in N$ is some element for which $g(n) = n''$. This is well-defined as if $n' \in N$ is another such element, $m \otimes n - m \otimes n' \in \mathrm{im}(\mathrm{id}_M \otimes f)$ by the exactness of

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \to 0$$

This is clearly a bilinear map, and so we have an induced map

$$\theta : M \otimes N'' \to M \otimes N / \mathrm{im}(\mathrm{id}_M \otimes f)$$

such that on pure tensors, the map is analogous to the one described above.

To see this is the inverse of θ, for $m \otimes n + \mathrm{im}(\mathrm{id}_M \otimes f)$

$$\theta'(\theta(m \otimes n + \mathrm{im}(\mathrm{id}_M \otimes f))) = \theta'(m \otimes g(n) = m \otimes n$$

and

$$\theta(\theta'(m \otimes n'')) = \theta(m \otimes n + \ker(\mathrm{id}_M \otimes f)) = m \otimes g(n) = m \otimes n''$$

Therefore, these maps are inverses on the generators, so inverses on the whole modules.

This shows that $M \otimes -$ is right exact. □

In general, these tensors functors are not exact. For this we would require $M \otimes -$ to preserve injectivity. When this is the case, we call M *flat*.

Definition 23. A right R-module M is called *flat* if

$$0 \to N' \xrightarrow{f} N$$

exact implies

$$0 \to M \otimes N' \xrightarrow{\mathrm{id}_M \otimes f} M \otimes N$$

is exact.

Proposition 5. *$M \otimes -$ is exact if and only if M is flat.*

Similarly, $- \otimes M$ is exact if and only if M is flat (where the definition of flat is generalized for left R-modules).

N.B. In the case R is commutative, if M is right R-module then we can put the structure of a left R-module on M by $r \cdot m = m \cdot r$. The same gives a right R-module action from a left.

In this case we can compute $M \otimes N$ for any type of R-modules M and N. Also, $M \otimes N$ has the structure of an R-module by

$$r \cdot (m \otimes n) = (r \cdot m) \otimes n = m \otimes (r \cdot n)$$

This makes $\otimes$ a bifunctor from $R - \underline{mod} \times R - \underline{mod} \to R - \underline{mod}$. This is a canonical example of a $\otimes$-*category*—a category $\mathcal{C}$ with a bi-functor $\otimes : \mathcal{C} \otimes \mathcal{C} \to \mathcal{C}$ satisfying some axioms.

2.3 Projective and Injective Modules

> "Did you think anymore about the product of two additive categories being additive?"
> "No."
> "Neither did I."
>
> Cogdell and Vilas

As we will see projective modules are generalization of free modules, we will first look at properties of free mdoules over a ring R.

Definition 24 (Free modules by Universal Mapping Property)**.** Let X be a set. The *free R-module generated by X* is an R-module F_X with an embedding

$$u : X \hookrightarrow F_X$$

such that for any set map $\phi : X \to M$ (M and R-module) there is a unique R-linear map $\widetilde{\phi} : F_X \to M$ such that $\phi = \widetilde{\phi} \cdot u$.

The construction of such an object is given by $F_X = \bigoplus_{x \in X} R$. We have $u : X \to F_X$ by $u(x) = (0, 0, \dots, 1, 0, \dots)$ where 1 occurs in the place corresponding to x. Given $\phi : X \to M$ the induced map is given by $\widetilde{\phi}(u(x)) = \phi(x)$ and then extended uniquely by linearity.

From the construction we see F_X is determined by the cardinality of X.

2.4 Free Modules

There are 4 basic properties of free modules:

1. Every module is a quotient of a free module.

   Proof. Take for X a generating set for M. Take $F = F_X$ with the embedding $\phi : X \hookrightarrow F_X$. By the universal property, there is a module morphism $\widetilde{\phi} : F_X \to M$. As M is generated by X, this morphism is surjective. □

2. Any morphism to a quotient can be lifted. This means

$$\begin{tikzcd} & F \arrow[d, "f"] \arrow[dl, dotted, "h"'] & \\ M \arrow[r, "g"'] & M' \arrow[r] & 0 \end{tikzcd}$$

   Proof. Let F be a free module in X. For $\bar{x} \in X$, $f(\bar{x}) = m''_X \in M''$. As g is surjective, we pull back to m_X with $g(m_X) = m''_X$. Define $\phi(\bar{x}) = m_X$. Then by the universal mapping property of F_X there is a module morphism $h = \widetilde{\phi}$, and this makes the diagram commute. □

3. The convariant Hom functor $\hom_R(F, -)$ is exact.

Proof. Begin with

$$0 \to M' \to M \xrightarrow{g} M'' \to 0$$

exact. As $\hom(F, -)$ is left exact, we need only check

$$\hom(F, M) \xrightarrow{\hom(F,g)} \hom(F, M'') \to 0$$

is surjective. But this exactly the fact that morphisms onto quotients can be lifted. □

4. Any short exact sequence ending in F splits.

   Proof. Let

   $$0 \to M' \to M \xrightarrow{g} F \to 0$$

   be exact. We want a morphism $s : F \to M$ so that $g \cdot s = \mathrm{id}_F$. Again this comes from 2—take $\mathrm{id} : F \to F$ and as $M \xrightarrow{g} F \to 0$ gives a quotient, we can lift g to a morphism $s : F \to M$. □

2.4.1 Projective Modules

We can generalize some the properties of free modules to general categories to get projective modules. The most important property of the previous 4 basic facts was that of lifting a map onto a quotient. As lifting can be defined just in terms of morphisms, this can be generalized to arbitrary categories.

Definition 25. An R-module P is *projective* if we have the following commutative diagram in R-<u>mod</u>

$$\begin{tikzcd} & P \arrow[dl, dotted, "h"'] \arrow[d, "f"] & \\ M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

There are 4 equivalent properties of projective modules.

Proposition 6. *The following are equivalent.*

1. *We have the following commutative diagram*

$$\begin{tikzcd} & P \arrow[dl, dotted, "h"'] \arrow[d, "f"] & \\ M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

2. *Every short exact sequence ending in P splits.*

3. *P is a direct summand of a free module. This means there exists N such that $N \oplus P = F$ is free.*

4. *The covariant Hom functor $\hom_R(P, -)$ is exact.*

Proof. $1 \Rightarrow 2$ Lift the identity morphism $\mathrm{id}_P : P \to P$ through $g : M \to P$.

$2 \Rightarrow 3$ As P is the quotient of a free module we have

$$0 \to N \to F \to P \to 0$$

exact. This sequence splits, so $F \cong N \oplus P$.

$3 \Rightarrow 1$ Write $F = N \oplus P$. Suppose we have a diagram

$$\begin{tikzcd} & P \arrow[d, "f"] & \\ M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

with the bottom row exact.

Then we have

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r] & F \arrow[r, "p"] \arrow[d, dotted, "h'"] & P \arrow[l, bend right, "s"'] \arrow[r] \arrow[d, "f"] & 0 \\ & & M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

From lifting $f \cdot p$ through g.

But we have a section $s : P \to F$, and the map $s \cdot h'$ is what we want.

$1 \Rightarrow 4$ This is exactly the same as in free modules: given

$$0 \to M' \to M \to M'' \to 0$$

we need only check

$$\hom(P, M) \to \hom(P, M'') \to 0$$

is surjective, which is the case from lifting.

$4 \Rightarrow 1$ This follows from $\hom(P, g)$ is surjective onto $\hom(P, M'')$—the lift is the preimage of $f : P \to M''$ under this map $\square$

From the fact that free modules are projective, and every module is the quotient of a free module, every module M is the quotient of a projective module. We usually phrase this as "R-<u>mod</u> has enough projectives"

This is important in homological algebra when we want to express things in terms of projective resolutions.

Proposition 7. *Every projective module P is flat.*

N.B. Note this uses exercise 19 in the homework: If I is an index set, then $N = \oplus_{\alpha \in I}$ is flat if and only if each N_α is flat for left R-modules.

Proof. We first cliam that R is flat as an R-module. Indeed, $N \otimes R \cong N$ so the image of an exact sequence is the same sequence.

Next, we claim that any free R-module is flat. This follows from the above note.

Finally, as P is the direct summand of a free module, the note above gives P is flat. $\square$

2.4.2 Aside: Exactness in abelian categories

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

In the case of modules, exactness at B means $\mathrm{im}(f) = \ker(g)$. For general categories, we then say exactness at B is $g \circ f = 0$ (because we have the notion of the zero map in abelian categories with the zero object) and if $k : K \to B$ is the kernel of g and $c : B \to F$ the cokernel of f, $c \circ k = 0$. The first condition is equivalent to $\mathrm{im}(f) \subset \ker(g)$, while the second is equivalent to $\ker(g) \subset \mathrm{im}(f)$.

2.5 Injective Modules

> "So that's what Folland looks like these days"
> "That's a book, not Folland."
>
> Cogdell and Nicolas Bolle

Injective modules historically preceeded projectives and are the categorical duals of projective modules and relate to the possibility of *dividing* by elements of R.

Definition 26. An R-module Q is *injective* if whenver we have a diagram that is exact on the top row

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M \\ & Q & \end{tikzcd}$$

We can extend f to a morphism

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M \arrow[dl, dotted, "h"] \\ & Q & \end{tikzcd}$$

making the diagram commutes.

One can think of this in the following way: M' sits inside of M as a submodule by the exactness of the top diagram, so Q injective means we can extend the map f to all of M.

Proposition 8. *The following are equivalent:*

1. *Q is injective*
2. *The contravariant Hom functor* $\hom(-, Q)$ *is exact*
3. *Any exact sequence*
$$0 \to Q \to M \to M' \to 0$$
splits

Proof. $1 \Leftrightarrow 2$: One can just dualize the statement for projectives, or: We have $\hom(-, Q)$ is left exact, thus we must show that given

$$0 \to M' \xrightarrow{f} M$$

exact we have $\hom(M, Q) \to \hom(M', Q)$ surjective. This follows immediately from the ability to lift morphisms given in the definition.

Likewise, the fact that we can lift is exactly that $\hom(M, Q) \to \hom(M', Q)$ is surjective.

$1 \Rightarrow 3$: Given

$$0 \to Q \to M \to M' \to 0$$

lift the morphism $\mathrm{id}_Q : Q \to Q$.

$3 \Rightarrow 1$: We first need the following:

Theorem 4. *Every module is the submodule of an injective.*

This is the analogue of projectives being a direct summand of a free module.

Proof. We utilize the connection between injectivity and divisibility. We first do the case of $\mathbb{Z}$-modules, in which case the notion of divisibility is a divisible abelian group (every map $t \mapsto mt$ for $m \in \mathbb{Z}$ non-zero is surjective).

Proposition 9. *If T is a divisible group then it is injective.*

Proof. Consider a diagram of abelian groups

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "i"] \arrow[d, "f"] & M \arrow[dl, dotted, "h"] \\ & T & \end{tikzcd}$$

where we want to construct the dotted morphism. For $x \in M$, $x \notin M'$, if x is free over M' ($\langle x \rangle \cap M' = \emptyset$) we can extend f to x by choosing $t \in T$ and defining $h(x) = t$. In the case x is not free, take n to be the smallest positive integer with $nx \in M'$. Then $f(nx)$ exists and is in T. As T is divisible, we can choose $u \in T$ with $nu = f(nx)$. We then define $h(x) = u$.

In this way, we can always expand h from f one element at a time, and it is easy to show this extension is well-defined.

Use Zorn's Lemma to construct a morphism g that extends f to a maximal subgroup of M (The poset will be a pair $(f_{M''}, M'')$ with $(f_{M''}, M'') \leq (f_H, H)$ if both $M'' \subseteq H$ and $f_{M''} = f_{H|M''}$—the maximal element of the chain is just the function on the union of all the sets, and the funtion taking an element to where any function in the chain (defined on the element) sends the element). Because we can extend any morphism on a proper subset of M, the maximally defined function must be defined on all of M.

This shows T is injective. □

We finish our claim for abelian groups.

Proposition 10. *Any abelian group is a subgroup of a divisible group.*

Proof. We must first define a notion of dual for an abelian group. Define $M^* = \hom_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$. If F is a free abelian group, than the dual is divisible. Indeed, any element on the dual is defined by the action of the function on its generators, and we can divide these images by any integer in $\mathbb{Q}/\mathbb{Z}$.

Now let M be an abelian group, and M^* its dual. Thus M^* is a quotient of a free group

$$F \to M^* \to 0$$

applying duality again yields

$$0 \to M^{**} \to F^*$$

an exact sequence. We claim $M \hookrightarrow M^{**}$ by the evaluation map. This is equivalent to saying that for every $m \in M$ with $m \neq 0$ there is a linear funtion $M \to \mathbb{Q}/\mathbb{Z}$ on which the image of m is non-zero. Again we utilize the divisibilty of $\mathbb{Q}/\mathbb{Z}$ to extend a map $\langle m \rangle \to \mathbb{Q}\mathbb{Z}$ to all of M. Thus we are left with finding a non-zero morphism $\langle m \rangle \to \mathbb{Q}/\mathbb{Z}$—iff m has torsion, send it to $\frac{1}{k} + \mathbb{Z}$ where k is the torsion, and if not, send it to $\frac{1}{2} + \mathbb{Z}$. Thus we have $M \hookrightarrow F^{**}$ where the latter group is divisible and thus injective. $\square$

Now we extend this to R-modules. There are 4 steps:

1. If R is any ring and A an abelian group, then $\hom_{\mathbb{Z}}(R, A)$ is naturally a left R module where the module structure is given by $r \cdot f(x) = f(xr)$[4].

2. If M is any R-module, there exists a natural isomorphism of abelian groups

$$\hom_{\mathbb{Z}}(M, A) \cong \hom_R(M, \hom_{\mathbb{Z}}(R, A))$$

given by $psi \mapsto f_\psi$ where $f_\psi(m)(r) = \psi(rm)$. The inverse is given by $f \mapsto \psi_f$ where $\psi_f(m) = f(m)(1)$. We interpret this as a natural isomorphism between $\hom_{\mathbb{Z}}(-, A)$ and $\hom_R(-, \hom_{\mathbb{Z}}(R, A))$.

3. If T is a divisible abelian group, then $\hom_{\mathbb{Z}}(R, T)$ is injective as an R-module. We show that $\hom_R(-, \hom_{\mathbb{Z}}(R, T))$ is an exact functor. Take an exact sequence

$$0 \to M' \to M \to M'' \to 0$$

and get

$$0 \to \hom_R(M'', \hom_{\mathbb{Z}}(R, T)) \to \hom_R(M, \hom_{\mathbb{Z}}(R, T)) \to \hom_R(M', \hom_{\mathbb{Z}}(R, T)) \to$$

with the left portion exact. By the natural isomorphism from the previous step

$$\begin{tikzcd}
0 \arrow[r] & \hom_R(M'', \hom_{\mathbb{Z}}(R, T)) \arrow[r] \arrow[d, leftrightarrow] & \hom_R(M, \hom_{\mathbb{Z}}(R, T)) \arrow[r] \arrow[d, leftrightarrow] & \hom_R(M', \hom_{\mathbb{Z}}(R, T)) \arrow[r] \arrow[d, leftrightarrow] & 0 \\
0 \arrow[r] & \hom_{\mathbb{Z}}(M'', T) \arrow[r] & \hom_{\mathbb{Z}}(M, T) \arrow[r] & \hom_{\mathbb{Z}}(M'', T) \arrow[r] & 0
\end{tikzcd}$$

The bottom sequence is exact because T is injective and hence we have surjectivity at $\hom_{\mathbb{Z}}(M'', T)$.

4. View M as an abelian group. Then

$$0 \to M \to T$$

with T divisible. Apply $\hom_{\mathbb{Z}}(R, -)$ to get

$$0 \to \hom_{\mathbb{Z}}(R, M) \to \hom_{\mathbb{Z}}(R, T)$$

which is a sequence in R modules. The right most term is injective by the previous part. As

$$M = \hom_R(R, M) \hookrightarrow \hom_{\mathbb{Z}}(R, M) \hookrightarrow \hom_{\mathbb{Z}}(R, T)$$

this proves the theorem.

[4] Note the multiplcation occurs on the right!

$\square$

With this theorem in hand, if we have a diagram with the top line exact

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' \\ & Q & \end{tikzcd}$$

We can put an exact sequence on the bottom with I injective

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' & & \\ 0 \arrow[r] & Q \arrow[r, shift left, "i"] & I \arrow[l, shift left, bend left, "t"] \arrow[r] & N'' \arrow[r] & 0 \end{tikzcd}$$

we can get a diagram by lifting $i \circ f$

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' \arrow[d, dotted, "h"] & & \\ 0 \arrow[r] & Q \arrow[r, shift left, "i"] & I \arrow[l, shift left, bend left, "t"] \arrow[r] & N'' \arrow[r] & 0 \end{tikzcd}$$

Giving us

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' \arrow[d, dotted, "h"] \arrow[dl, dotted, "t \circ h"'] & & \\ 0 \arrow[r] & Q \arrow[r, shift left, "i"] & I \arrow[l, shift left, bend left, "t"] \arrow[r] & N'' \arrow[r] & 0 \end{tikzcd}$$

For $m \in M'$ we then have $i(f(m)) = h(g(m))$, so $f(m) = t(i(f(m))) = t(h(g(m)))$ which gives us the lift we need. $\square$

3 Homological Algebra

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While homological algebra was formally introduced in 1940, there are several earlier theorems that use the basic techniques of homological algebra. Some examples are Hilbert's 1890 Syzygy Theorem, Schur's 1904 work on projective representations of groups ($H^2(G, \mathcal{C}^\times)$), Schreier's 1926 work in group extensions ($H^2(G, A)$), and Noether's 1929 theory of cssed product algebras ($H^2(G, E)$).

While this area can be done in the general setting of abelian categories, we will restrict ourselves to R-modules.

3.1 Chain Complexes

Definition 27. A *chain (or descending) complex* $C_\bullet$ is an index set $\{C_i, d_i\}_{i\in\mathbb{Z}}$ of R-modules C_i with module morphisms $d_i : C_i \to C_{i-1}$ such that $d_{i-1} \circ d_i = 0$.

Example 21.

1. If M is an R-module, we can form a complex with $C_0 = M$ and everything else being the 0 object or map.
2. Given a short exact sequence in $R - \underline{mod}$

$$0 \to M' \xrightarrow{f} M \xrightarrow{g} M' \to 0$$

we can extend this to an exact chain by extending with the 0 object.

3. *The Projective Resolution of a Module*: Given a module M, M is the quotient of a projective module. $0 \to N_0 \to P_0 \to M \to 0$. Likewise, N_0 is the quotient of a projective module, so

$$0 \to N_1 \to P_1 \to N_0 \to 0$$

Continuing in this way we obtain a chain complex

$$\cdots \to P_2 \to P_1 \to P_0 \to M \to 0 \to \cdots$$

where the map $P_{i+1} \to P_i$ is given by $P_{i+1} \to N_i \to P_i$.

4. *Hilbert's Syzygy*[5] *Theorem*: If M is a module and X is a generating set for M, then

$$0 \to N_0 \to F_X = F_0 \to M \to 0$$

is an exact sequence. In this case, N_0 acts as relations amog the generators of M. If then Y is a generating set for N_0 we get another exact sequence

$$0 \to N_1 \to F_Y = F_1 \to N_0 \to 0$$

Continuitng we get our chain

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0 \to \cdots$$

where $F_{i+1} \to F_i$ factors through N_i.

The Syzygy Theorem states if M is a module for $F[x_1, \ldots, x_n]$, then this free resolution of M has at most m steps. Thus, the relations among the relations among the relations etc. eventually become trivial.

3.2 Morphisms of Complexes

Definition 28. Given two complexes $C_\bullet = \{C_i, d_i\}$ and $C'_\bullet - \{C'_i, d'_i\}$ a *morphism of complexes* $\alpha : C_\bullet \to C'_\bullet$ consists of an indexed set of morphisms $\alpha_i : C_i \to C'_i$ such that each box commutes

$$\begin{tikzcd}
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\alpha_{i+1}"] & C_i \arrow[r, "d_{i+1}"] \arrow[d, "\alpha_i"] & Ci-1 \arrow[r] \arrow[d, "\alpha_{i-1}"] & \cdots \\
\cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"] & C'_i \arrow[r, "d'_{i+1}"] & C'i-1 \arrow[r] & \cdots
\end{tikzcd}$$

[5]The term syzygy comes from Hilbert's notation that the *syzygy* module of a module M is the module of relations between generators

This is usually written as $\alpha d = d'\alpha$.

This allows us to create $R - \underline{comp}$, the (abelian) category of chain complexes. In fact $R - \underline{mod}$ is a full subcategory of $R - \underline{comp}$ by identifying a module M with the chain complex consisting of M in one coordinate and 0 everywhere else.

If $C_\bullet, C'_\bullet$ are two complexes we can form the direct sum $C_\bullet \oplus C'_\bullet = \{C_i \oplus C'_i, d_i \oplus d'_i\}$. One can extend this to the direct sum of a family of complexes.

3.3 Homology

Let $C_\bullet$ be a complex

$$\cdots \to C_{i+1} \to C_i \to C_{i-1} \to \cdots$$

to measure exactness, we can look at $\ker d_i / \operatorname{im} d_{i+1}$. If the complex is exact at C_i, this module is 0.

Definition 29. The *cycle modules* are $Z_i = Z_i(C_\bullet) = \ker d_i$, and the *boundary modules* are $B_i = B_i(C_\bullet) = \operatorname{im} d_{i+1}$. The *homology modules* are $H_i = H_i(C_\bullet) = Z_i/B_i$.

Suppose $\alpha : C_\bullet \to C'_\bullet$ is a morphism of complexes

$$\begin{tikzcd}
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\alpha_{i+1}"] & C_i \arrow[r, "d_i"] \arrow[d, "\alpha_i"] & C_{i-1} \arrow[r] \arrow[d, "\alpha_{i-1}"] & \cdots \\
\cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"] & C'_i \arrow[r, "d'_i"] & C'_{i-1} \arrow[r] & \cdots
\end{tikzcd}$$

Then we see $\alpha(Z_i) \subset Z'_i$ and $\alpha_i(B_i) \subset B'_i$. We have induced maps

$$H_i(C_\bullet) = Z_i/B_i \xrightarrow{\tilde{\alpha}_i} Z'_i/B'_i = H_i(C'_\bullet)$$

by

$$z + B_i \mapsto \alpha_i(z) + B'_i$$

The well-definedness comes from the fact that $\alpha(Z_i) \subset Z'_i$ and $\alpha_i(B_i) \subset B'_i$.

As $\tilde{\alpha}_i \in \hom(H_i(C_\bullet), H_i(C'_\bullet))$, H_i is a covariant functor $H_i : R - \underline{comp} \to R - \underline{comp}$. Moreover, these functors are additive. Therefore $H_i(C_\bullet \oplus C'_\bullet) = H_i(C_\bullet) \oplus H_i(C'_\bullet)$.

3.4 Cohomology

Cohomology is defined in an analogous way to homology, except the arrows are reversed.

Definition 30. A *cochain complex (or ascending complex)* is a sequence of R-mods $C^\bullet = \{C^i, d^i\}$ where $d^i : C^i \to C^{i+1}$, $d^2 = 0$.

N.B. Note that any chain complex $C_\bullet$ can be made into a cochain complex $C^\bullet$ by $C_i \mapsto C^{-i}$.

Morphisms of a cochain complex are sequences of R-morphism $\alpha_i : C^i \to C'^i$ making the following diagram commute

$$\begin{tikzcd}
\cdots \arrow[r] & C^{i-1} \arrow[r, "d^{i-1}"] \arrow[d, "\alpha_i"] & C^i \arrow[r, "d^i"] \arrow[d, "\alpha_i"] & C^{i+1} \arrow[r, "d^{i+1}"] \arrow[d, "\alpha_{i+1}"] & \cdots \\
\cdots \arrow[r] & C'^{i-1} \arrow[r, "d'^{i-1}"] & C'^i \arrow[r, "d'^i"] & C'^{i+1} \arrow[r, "d'^{i+1}"] & \cdots
\end{tikzcd}$$

Then we define homology by $Z^i(C_\bullet) = \ker d^i$, $B^i(C_\bullet) = \operatorname{im} d^{i-1}$ and $H^i(C_\bullet) = Z^i/B^i$.

Then in an analogous way to homology, we see morphisms on cochain complexes induce a morphism on cohomology making H^i covariant, additive functors $R - \underline{cocomplex} \to R - \underline{mod}$.

3.5 Exact sequences of Complexes

Because cochain complexes and chain complexes are virtually the same, we will just focus on chain complexes and homology.

We can define the notion of a (short) exact sequence of chain complexes. Given chain complexes $A_\bullet, B_\bullet, C_\bullet$ with morphisms $\alpha : A_\bullet \to B_\bullet$ and $\beta : B_\bullet \to C_\bullet$, the commuting diagram

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & A_{i+1} \arrow[r, "d_A"] \arrow[d, "\alpha"] & A_i \arrow[r] \arrow[d] & A_{i-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & B_{i+1} \arrow[r, "d_B"] \arrow[d, "\beta"] & B_i \arrow[r] \arrow[d] & B_{i-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_C"] \arrow[d] & C_i \arrow[r] \arrow[d] & C_{i-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

is exact if each of the columns are. From this, we get induced maps on the homology

$$H_i(A) \to H_i(B) \to H_i(C)$$

In fact, using a connecting morphism (which we will develop below) we actually can convert this to a long exact sequence of homology

$$\cdots \to H_{i+1}(A) \to H_{i+1}(B) \to H_{i+1}(C) \xrightarrow{\delta} H_i(A) \to \cdots$$

3.6 The Snake Lemma

In order to get the connecting morphism, we first develop the following lemma by way of a diagram chase.

Lemma 3 (Snake Lemma)**.** *Suppose we are given a diagram*

$$\begin{tikzcd}
 & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N' \arrow[r] & N \arrow[r] & N''
\end{tikzcd}$$

with exact rows.

Then we get an induced diagram

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
 & \ker d' \arrow[r] \arrow[d] & \ker d, \arrow[r] \arrow[d] & \ker d'' \arrow[d, "i"] \arrow[dddll, rounded corners] & \\
 & M' \arrow[r, "f"] \arrow[d, "d'"] & M \arrow[r, "g"] \arrow[d, "d"] & M'' \arrow[r] \arrow[d, "d''"] & 0 \\
0 \arrow[r] \arrow[dd, rounded corners] & N' \arrow[r, "h"] \arrow[d, "\pi"] \arrow[uu, rounded corners, to path={-- ([xshift=2ex]\tikztostart.east) |- ([yshift=-2ex]\tikztotarget.south) -- (\tikztotarget)}, "\delta"] & N \arrow[r, "k"] \arrow[d] & N'' \arrow[d] & \\
 & \operatorname{coker} d' \arrow[r] \arrow[d] & \operatorname{coker} d \arrow[r] \arrow[d] & \operatorname{coker} d'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

So that the sequence

$$\cdots \to \ker d' \to \ker d \to \ker d'' \xrightarrow{\delta} \operatorname{coker} d' \to \operatorname{coker} d \to \operatorname{coker} d'' \to \cdots$$

is exact.

Proof. For $x \in \ker d''$, then choose $y \in M$ so that $g(y) = i(x)$ [this is possible as g surjects onto M'']. Note that $k(d(y)) = d''(g(y)) = d''(i(x)) = 0$. Therefore $d(y) \in \operatorname{im} h$, so we can define $\delta(x) = \pi(h^{-1}(d(y)))$.

It must be checked that this is well-defined. The only point of contention is in the choice of y. If y' is another choice, then $y - y' \in \ker g$, so $f^{-1}(y - y') \in M'$, and we see that $\pi(h^{-1}(d(y - y'))) = \pi(f^{-1}(y - y')) = 0$, so the δ defined based on y or y' is the same, whence δ is well-defined. This is also easily seen to be a homomorphism, as $g(a + b) = g(a) + g(b)$, so if we choose $a \in g^{-1}(x)$ and $b \in g^{-1}(y)$ to define $\delta(x), \delta(y)$ we can choose $ra + b$ to define $\delta(rx + y)$. As the rest of the definition of δ is the composition of morphisms, we have that δ is a morphism. $\square$

3.7 Long Sequence of Homology

We can now apply the Snake Lemma in the setting of homology.

By the definitions of cycles and boundaries we get an exact sequence of complexes

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
 & Z'_i \arrow[r, "\alpha"] \arrow[d] & Z_i \arrow[r, "\beta"] \arrow[d] & Z''_i \arrow[d, "i"] \arrow[dd, bend left=60] & \\
 & C'_i \arrow[r, "\alpha_i"] \arrow[d, "d'_i"] & C_i \arrow[r, "\beta_i"] \arrow[d, "d_i"] & C''_i \arrow[r] \arrow[d, "d''_i"] & 0 \\
0 \arrow[r] \arrow[dd, bend right=60] & C'_{i-1} \arrow[r, "\alpha_{i-1}"] \arrow[d, "\pi"] & C_{i-1} \arrow[r, "\beta_{i-1}"] \arrow[d] \arrow[ll, bend right=60, "\delta"'] & C''_{i-1} \arrow[d] & \\
 & C'_{i-1}/B'_{i-1} \arrow[r, "\alpha_{i-1}"] \arrow[d] & C_{i-1}/B_{i-1} \arrow[r, "\beta_{i-1}"] \arrow[d] & C''_{i-1}/B''_{i-1} \arrow[d] & \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

By the Snake lemma we get a map $\delta : Z''_i \to C'_{i-1}/B'_{i-1}$. We now need to show that this map can factor through homology so that we can map $Z''_i/B''_i \to Z'_{i-1}/B'_{i-1}$ by $z/B''_i \mapsto \delta(z)$. In order for this to be well-defined, we must show that $\delta(B''_i) = 0$ and $\operatorname{im}\delta \subset Z'_{i-1}/B'_{i-1}$.

To check the first statement, if $x \in B''_i$, then we can write $x = d''(\xi)$ for some $\xi \in C''_{i+1}$. Then definition of δ leaves us with finding $y \in C_i$ with $\beta_i(y) = x$. But our choice can then be $d(y')$ where $y' \in \beta_{i+1}^{-1}(\xi)$. Then the next step of the Snake lemma sends $d(y') \mapsto d(d(y')) = 0$, so in the end $\delta(x) = 0$.

Now we check the last statement. Say $x \in Z''_i$ then for $y \in \beta_i^{-1}(x)$ we have $\delta(x) = \alpha_{i-1}^{-1}(d_i(y))$, so $d'(\delta(x)) = d'(\alpha_{i-1}^{-1}(d_i(y)))$. We claim this is 0. As α_{i-2} is injective, it suffices to show $\alpha_{i-2}(d'(\alpha_{i-1}^{-1}(d_i(y)))) = 0$. By the commutativity of the diagram

$$\alpha_{i-2}(d'(\alpha_{i-1}^{-1}(d_i(y)))) = d_{i-1}(\alpha_{i-1}(\alpha_{i-1}^{-1}(d_i(y)))) = d^2(y) = 0$$

Therefore δ does induce a map on homology. We are left to check exactness of this sequence.

Thus, given an exact sequence of complexes

$$0 \to C'_\bullet \xrightarrow{\alpha} C_\bullet \xrightarrow{\beta} C''_\bullet \to 0$$

the Snake Lemma gies a connecting morphism

$$\delta_i : H_i(C''_\bullet) \to H_{i-1}(C'_\bullet)$$

so that

$$\cdots \to H_i(C'_\bullet) \to H_i(C_\bullet) \to H_i(C''_\bullet) \xrightarrow{\delta_i} H_{i-1}(C'_\bullet) \to \cdots$$

is an exact sequence.

Now if we have a commutative diagram with exact rows

$$
\begin{tikzcd}
0 \arrow[r] & C'_\bullet \arrow[r, "\alpha"] \arrow[d, "\alpha"] & C_\bullet \arrow[r, "\beta"] \arrow[d, "f"] & C''_\bullet \arrow[r] \arrow[d, "f''"] & 0 \\
0 \arrow[r] & D'_\bullet \arrow[r, "\gamma"] & D_\bullet \arrow[r, "\kappa"] & D''_\bullet \arrow[r] & 0
\end{tikzcd}
$$

then for each i, we have the following commutative square

$$\begin{tikzcd} H_i(C''_\bullet) \arrow[r, "\delta_i^c"] \arrow[d, "\tilde{f}''_i"] & H_{i-1}(C'_\bullet) \arrow[d, "\tilde{f}'_{i-1}"] \\ H_i(D''_\bullet) \arrow[r, "\delta_i^D"] & H_{i-1}(D'_\bullet) \end{tikzcd}$$

Likewise, we can dualize the snake lemma to get a connecting homomorphism on cohomology.

$$\delta_i : H_i(C''_\bullet) \to H_{i+1}(C'_\bullet)$$

so that

$$\cdots \to H_i(C'_\bullet) \to H_i(C_\bullet) \to H_i(C''_\bullet) \xrightarrow{\delta_i} H_{i+1}(C'_\bullet) \to \cdots$$

is an exact sequence.

Now if we have a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & C'_\bullet \arrow[r, "\alpha"] \arrow[d, "\alpha"] & C_\bullet \arrow[r, "\beta"] \arrow[d, "f"] & C''_\bullet \arrow[r] \arrow[d, "f''"] & 0 \\ 0 \arrow[r] & D'_\bullet \arrow[r, "\gamma"] & D_\bullet \arrow[r, "\kappa"] & D''_\bullet \arrow[r] & 0 \end{tikzcd}$$

then for each i, we have the following commutative square

$$\begin{tikzcd} H_i(C''_\bullet) \arrow[r, "\delta_i^c"] \arrow[d, "\tilde{f}''_i"] & H_{i+1}(C'_\bullet) \arrow[d, "\tilde{f}'_{i+1}"] \\ H_i(D''_\bullet) \arrow[r, "\delta_i^D"] & H_{i+1}(D'_\bullet) \end{tikzcd}$$

3.8 Homotopies

Given two complexes $C_\bullet, C'_\bullet$ and a morphism $\alpha : C_\bullet \to C'_\bullet$ we get an induced map on homology

$$\tilde{\alpha}_i : H_i(C) \to H_i(C')$$

When do two maps $\alpha, \beta : C_\bullet \to C'_\bullet$ induce the same map on homology? One sufficient condition is that the maps are *homotopic*.

Definition 31. Let $\alpha, \beta : C_\bullet \to C'_\bullet$ be morphisms of complexes. We say α is *homotopic* to β ($\alpha \sim \beta$) if there is a sequence of morphisms $\{si\}$ where $s_i : C_i \to C'_{i+1}$ such that $\alpha_i - \beta_i = d'_{i+1}s_i + s_{i-1}d_i$

$$\begin{tikzcd} \cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\beta_{i+1}"', "\alpha_{i+1}"] & C_i \arrow[r, "d_i"] \arrow[d, "\beta_i"', "\alpha_i"] \arrow[dl, "s_i"] & C_{i-1} \arrow[r] \arrow[d, "\beta_{i-1}"', "\alpha_{i-1}"] \arrow[dl, "s_{i-1}"] & \cdots \\ \cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"'] & C'_i \arrow[r, "d'_i"'] & C'_{i-1} \arrow[r] & \cdots \end{tikzcd}$$

Proposition 11. *If $\alpha, \beta : C_\bullet \to C'_\bullet$ and $\alpha \sim \beta$, then $\tilde{\alpha}_i = \tilde{\beta}_i$ for all i.*

Proof. Recall that the induced map on homology is given by $\tilde{\alpha}_i(z_i + B_i) = \alpha_i(z_i) + B'_i$ and $\tilde{\beta}_i(z_i + B'_i) = \beta_i(z_i) + B'_i$.

Thus, the maps are the same if $\beta_i(z_i) - \alpha_i(z_i) \in B'_i$. But by the homotopy,

$$\beta_i(z_i) - \alpha_i(z_i) = d'_{i+1}s_i(z_i) + s_{i-1}d_i(z_i) = d'_{i+1}s_i(z_i) \in B'_i$$

□

Proposition 12. *Homotopy is an equivalence relation on morphisms of complexes*

Proof. This is clearly reflexive and symmetric. For transitivity, just add the morphisms. □

Proposition 13. *Homotopies can be multiplied (by composition), i.e., if $\alpha, \beta : C_\bullet \to C'_\bullet$ are homotopic and $\gamma, \delta : C'_\bullet \to C''_\bullet$ are homotopic, then $\gamma \circ \alpha \sim \delta \circ \beta$.*

Proof. Let s be the homotopy between $\alpha \sim \beta$ and t between $\gamma \sim \delta$.

We have the diagram

$$\begin{tikzcd}
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d] & C_i \arrow[r, "d_i"] \arrow[d, "\alpha_i"] \arrow[dl, "s_i"', "\beta_i"] & C_{i-1} \arrow[r] \arrow[d] \arrow[dl, "s_{i-1}"'] & \cdots \\
\cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"'] \arrow[d] & C'_i \arrow[r, "d'_i"'] \arrow[d, "\gamma_i"] \arrow[dl, "t_i"', "\delta i"] & C_{i-1} \arrow[r] \arrow[d] \arrow[dl, "t_{i-1}"'] & \cdots \\
\cdots \arrow[r] & C''_{i+1} \arrow[r, "d'_{i+1}"'] & C''_i \arrow[r, "d'_i"'] & C''_{i-1} \arrow[r] & \cdots
\end{tikzcd}$$

Now consider $\gamma_i(\alpha_i - \beta_i) = \gamma_i\alpha_i - \gamma_i\beta_i$ and $(\gamma_i - \delta_i)\beta_i = \gamma_i\beta_i - \delta_i\beta_i$. Then

$$\gamma_i(\alpha_i - \beta_i) = \gamma_i d'_{i+1} s_i + \gamma_i s_{i-1} d_i = d''_{i+1}\gamma_{i+1} s_i + \gamma_i s_{i-1} d_i$$

and

$$(\gamma_i - \delta_i)\beta_i = d''_{i+1} t_i \beta_i + t_{i-1} d'_i \beta_i$$

adding these yields

$$\gamma_i\alpha_i - \delta_i\beta_i = d''_{i+1}(\gamma_{i+1}s_i + t_i\beta_i) + (\gamma_i s_{i-1} + t_{i-1}\beta_{i-1})d_i$$

so our homotopies should be given as $\gamma_{i+1}s_i + t_i\beta_i$. □

3.9 Resolutions

"Oh I am not happy"

Cogdell

Begin with an R-module M. Then a *resolution* of M is a non-negative[6] complex $C_\bullet$ and an augmentation morphism $C_0 \xrightarrow{\epsilon} M \to 0$ so that

$$\to C_2 \to C_1 \to C_0 \xrightarrow{\epsilon} M \to 0 \to 0$$

is exact.

Definition 32. A resolution is *projective* if all P_i are projective for $i \geq 0$.

We have construted these projective resolutions before but we've never checked if they were exact. Recall the construction is given by first expressing a module as a quotient of a projective

$$0 \to N_0 \to P_0 \xrightarrow{\epsilon} M \to 0$$

then doing the same for N_i

$$0 \to N_{i+1} \to P_{i+1} \to N_i \to 0$$

and we create the sequence

[6] This means $C_{-i} = 0$ for i positive

$$\begin{tikzcd}
\cdots \arrow[r] & P_3 \arrow[r, "d_3"] & P_2 \arrow[rr, "d_2"] \arrow[dr] & & P_1 \arrow[rr, "d_1"] \arrow[dr] & & P_0 \arrow[r, "\epsilon"] & M \arrow[r] & 0 \\
& & & N_1 \arrow[ur] & & N_0 \arrow[ur] & & &
\end{tikzcd}$$

We check that this is an exact sequence, which follows easily from the exactness from the short exact sequences.

While we have the existence of projective and injective resolutions, we need some results about their uniqueness.

Theorem 5. *Let $(P_\bullet, \epsilon)$ be a projective resolution of M, and $(C'_\bullet, \epsilon')$ be any resolution of M'. Let $\mu : M \to M'$ be a module morphism. Then there is a morphism $\alpha : (P_\bullet, \epsilon) \to (C'_\bullet, \epsilon')$ so that $\mu\epsilon = \epsilon'\alpha_0$, i.e. the square containing the augmentations commute.*

$$\begin{tikzcd}
\arrow[r] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & 0 \\
\arrow[r] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & 0
\end{tikzcd}$$

Moreover, any two such morphisms $\alpha, \beta : (P_\bullet, \epsilon) \to (C'_\bullet, \epsilon')$ satisfying these conditions are homotopic.

Proof. The first assertion comes by way of induction: First, using the projectivity of P_0 we get a morphism $\alpha_0 : P_0 \to C'_0$ so that $\epsilon' \circ \alpha_0 = \mu \circ \epsilon$.

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r, "d_2"] & P_1 \arrow[r, "d_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
\cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

Now for the induction step, suppose we have constructed $\alpha_{n-1}, \cdots, \alpha_0$ so that the requisite commutivity holds.

$$\begin{tikzcd}
P_n \arrow[r, "d_n"] & P_{n-1} \arrow[r, "d_{n-1}"] \arrow[d, "\alpha_{n-1}"] & P_{n-2} \arrow[r, "d_{n-2}"] \arrow[d, "\alpha_{n-2}"] & \cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, "\alpha_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "\alpha_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
C'_n \arrow[r, "d'_n"] & C'_{n-1} \arrow[r, "d'_{n-1}"] & C'_{n-2} \arrow[r, "d'_{n-2}"] & \cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

Note that $d'_{n-1} \circ \alpha_{n-1} \circ d_n = \alpha_{n-2} \circ d_{n-1} \circ d_n$. Thus, $\operatorname{im} \alpha_{n-1} \circ d_n \subseteq \operatorname{im} d'_n$. Thus, we can restrict C'_{n-1} to $\operatorname{im} C'_n$ and lift the morphism $\alpha_{n-1} \circ d_n : P_n \to C'_{n-1}$ using the projectivity of P_n to C'_n, and this makes the diagram commute.

$$\begin{tikzcd}
P_n \arrow[r] \arrow[d, "\alpha_n"] & P_{n-1} \arrow[r] \arrow[d, "\alpha_{n-1}"] & P_{n-2} \arrow[r] \arrow[d, "\alpha_{n-2}"] & \cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, "\alpha_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "\alpha_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
C'_n \arrow[r] & C'_{n-1} \arrow[r] & C'_{n-2} \arrow[r] & \cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

By induction we thus get our morphism α.

Now we show this morphism is unique up to homotopy which we now construct. Define $\gamma = \alpha - \beta$. Then it suffices to construct a homotopy $\gamma \sim 0$. We again use induction. Note that $\epsilon'\gamma_0 = 0$ as α, β both satisfy commutivity with the augmentation map. In the base case we have the diagram

$$\begin{tikzcd}
 & P_0 \arrow[d, "\gamma_0"] & & \\
C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & 0
\end{tikzcd}$$

Again, as $\operatorname{im}\gamma_0 \subseteq \ker\epsilon'$, we can restrict C'_0 to $\operatorname{im} d'_1$ and lift γ_0 restricted to this codomain to get $s_0 : P_0 \to C'_1$. This makes $d'_1 \circ s_0 = \gamma_0$.

Now by induction if we have s_{n-1} we get

$$\begin{tikzcd}
P_n \arrow[r] \arrow[d, "\gamma n"] & P_{n-1} \arrow[r] \arrow[d, "\gamma n-1"] \arrow[dl, "s_{n-1}"] & P_{n-2} \arrow[r] \arrow[d, "\gamma_{n-2}"] \arrow[dl, "s_{n-2}"] & \cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, "\gamma_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "\gamma_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\gamma_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
C'_n \arrow[r] & C'_{n-1} \arrow[r] & C'_{n-2} \arrow[r] & \cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

so that $d'_{i+1}s_i = s_{i-1}d_i$. Consider the map $j : P_n \to C'_n$ by $x \mapsto \gamma_n(x) - s_{n-1}(d_n(x))$. Again, we can see $\operatorname{im} j \subseteq \operatorname{im} d'_{n+1} = \ker d'_n$ as

$$d'_n(\gamma_n - s_{n-1}d_n) = d'_n\gamma_n - d'_n s_{n-1}d_n = \gamma_{n-1}d_n - d'_n s_{n-1}d_n = (\gamma_{n-1} - d'_n s_{n-1})d_n = s_{n-2}d_{n-1}d_n = 0$$

so we can restrict j to the codomain $\operatorname{im} d'_{i+1}$ and then lift j to $s_n : P_n \to C'_{n+1}$. The map s_n then satisfies

$$d'_{n+1} \circ s_n = \gamma_n - s_{n-1}d_n$$

which is the condition we need for a homotopy. □

Corollary 1. *Suppose we have two projective resolutions $(P_\bullet, \epsilon)$ and $(P'_\bullet, \epsilon')$ for a given module M. Then there is a morphism $\alpha : P_\bullet \to P'_\bullet$ lifting id_M. This is unique up to homotopy. Now if $\beta : P'_\bullet \to P_\bullet$ lifts id_M, then α, β are inverses up to homotopy.*

Proof. Consider

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow P \longrightarrow 0$$

$$\cdots \longrightarrow P'_1 \longrightarrow P'_0 \longrightarrow P' \longrightarrow 0$$

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow P \longrightarrow 0$$

$$\cdots \longrightarrow P'_1 \longrightarrow P'_0 \longrightarrow P' \longrightarrow 0$$

□

Definition 33. A morphism $\alpha : C_\bullet \to C'_\bullet$ is an *quasi-isomorphism* if the induced maps

$$\tilde{\alpha} : H_i(C_\bullet) \to H_i(C'_\bullet)$$

on homology are isomorphisms.

Now consider the resolutions

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r] & P_1 \arrow[r] & P_0 \arrow[r] \arrow[d, "\epsilon"] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & M \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

This is a quasi-isomorphism, as $H_i(P_\bullet) = 0$ for $i \geq 1$ and $H_0(P_\bullet) = \ker(0)/\operatorname{im} d_1 \cong M$.

3.10 Injective Resolutions

Definition 34. If M is an R-module, a *corresolution* of M is a non-negative cochain complex $C^\bullet$ along with a *coaugmentation morphism* $\eta : M \to C^0$ such that

$$\to 0 \to M \xrightarrow{\eta} C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} \cdots$$

is exact.

In the case all C^i are injective, this is called an *injective resolution.*

As R-<u>mod</u> has enough injectives, we can construct an injective resolution for any module M: take

$$0 \to M \to Q^0 \to N^0 \to 0$$

with Q^0 injective, and continue with

$$0 \to N^0 \to Q^1 \to N^1 \to 0$$

and combine them into a cochain complex.

CHANGE

$$\begin{tikzcd}
\cdots \arrow[r] & M \arrow[r, "d_0"] & Q^0 \arrow[rr, "d^0"] \arrow[dr] & & Q^1 \arrow[rr, "d_1"] \arrow[dr] & & Q^0 \arrow[r, "\epsilon"] & M \arrow[r] & 0 \\
& & & N^1 \arrow[ur] & & N^0 \arrow[ur] & & &
\end{tikzcd}$$

The analogues of the projective resolutions follow by way of duality for injective resolutions.

Theorem 6. *Let $(Q^\bullet, \eta)$ be an injective resolution of M and $(C'^\bullet, \eta')$ any co-resolution of M'. Then if $\lambda' : M' \to M$ is any module morphism, there is a morphism $\beta : C'^\bullet \to Q^\bullet$ such that $\beta^0 \circ \eta' = \eta \circ \lambda'$. Moreover, any two such morphisms β, β' are homotopic.*

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r, "\eta'"] \arrow[d, "\lambda'"] & C'^0 \arrow[r, "d'^0"] \arrow[d, "\beta^0"] & C'^1 \arrow[r, "d'^1"] \arrow[d, "\beta^1"] & C'^2 \arrow[r] \arrow[d, "\beta^2"] & \cdots \\
0 \arrow[r] & M \arrow[r, "\eta"] & Q^0 \arrow[r] & Q^1 \arrow[r] & Q^2 \arrow[r] & \cdots
\end{tikzcd}$$

Theorem 7. *If $(Q^\bullet, \eta)$ and $(I^\bullet, \eta')$ are two injective resolutions of M. Then there is a morphism $\beta : I^\bullet \to Q^\bullet$ which lifts id_M, and this lift is unique up to homotopy.*

This basically says the injective resolutions of a module are unique up to homotopy.

N.B. These injective resolutions are used to define either right derived functors for covariant functors or left derived functors for contravariant functors.

N.B. This ends the material for the first exam.

3.11 Derived Functors

Suppose $F : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ is a covariant functor. For the most part we will deal with $F = \hom(M, -)$ which will give us Ext, and $F = M \otimes -$ which will give us Tor. We also assume that F is additive.

The idea is as follows: take a module M and replace it with a projective resolution $(P_\bullet, \epsilon)$ of M.

$$\cdots \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \xrightarrow{\epsilon} M \longrightarrow 0$$

Then applying the functor F yields.

$$\cdots \longrightarrow F(P_2) \longrightarrow F(P_1) \longrightarrow F(P_0) \xrightarrow{F(\epsilon)} F(M) \longrightarrow 0$$

This remains a complex.

In the case F is exact, this remains a resolution of $F(M)$. We can compute its homology H_i. As F is exact, $H_i = 0$. If F is not exact, the homology becomes more interesting. This is how we get the derived functors.

Definition 35. The *left derived functor* $L_nF(M) = H_n(F(P_\bullet))$ for $n \geq 0$.

In particular, we have $L_0(F(M)) = \operatorname{coker}(F(d_1)) = F(P_0)/F(d_1)F(P_n) = F(P_0)/B_1(F(P_\bullet))$.

Recall that as projective resolutions are unique up to homotopy and therefore homology, these derived functors are well-defined up to isomoorphism.

How does this functor act on morphisms? Suppose $\mu : M \to M'$ is a morphism. We need a map $L_nF(\mu) : L_n(F(M)) \to L_n(F(M'))$. By our above work, if we have a morphism μ and projective resolutions of M, M', this induces a map of complexes $\alpha : (P_\bullet, \epsilon) \to (P'_\bullet, \epsilon')$.

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r] \arrow[d, "\alpha_2"] & P_1 \arrow[r] \arrow[d, "\alpha_1"] & P_0 \arrow[r] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & 0 \\
\cdots \arrow[r] & P'_2 \arrow[r] & P'_1 \arrow[r] & P'_0 \arrow[r] & M' \arrow[r] & 0
\end{tikzcd}$$

We can then apply F to all of this to get

$$\begin{tikzcd}
\cdots \arrow[r] & F(P_2) \arrow[r] \arrow[d, "F(\alpha_2)"] & F(P_1) \arrow[r] \arrow[d, "F(\alpha_1)"] & F(P_0) \arrow[r] \arrow[d, "F(\alpha_0)"] & F(M) \arrow[r] \arrow[d, "F(\mu)"] & 0 \\
\cdots \arrow[r] & F(P'_2) \arrow[r] & F(P'_1) \arrow[r] & F(P'_0) \arrow[r] & F(M') \arrow[r] & 0
\end{tikzcd}$$

Note that we still have $F(\epsilon') \circ F(\alpha_0) = F(\mu) \circ F(\epsilon)$ and moreover all the other squares commute. Thus we have induced maps on homology. These induced maps are $L_n(\mu) = \widetilde{F(\alpha_n)}$. As our lifts α are unique up to homotopy, these functions are well-defined up to isomorphism, as F respects this homotopy.

3.12 Connecting Homomorphisms

We have that the left derived functors are homology functors for projective resolutions. Similarly, given a short exact sequence of complexes

$$0 \longrightarrow C'_\bullet \longrightarrow C_\bullet \longrightarrow C''_\bullet \longrightarrow 0$$

we get a long exact sequence of homology functors by the Snake Lemma.

Does this induce a long exact sequence of derived functors when given an exact sequence of modules

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

The only way this could happen is if when we replace the modules with their projective resolutions we get a short exact sequence of resolutions.

Definition 36. By a *projective resolution of a short exact sequence of modules*

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

we mean projective resolution of each module $(P'_\bullet, \epsilon')$ of M', $(P_\bullet, \epsilon)$ of M and $(P''_\bullet, \epsilon'')$ of M'' together with chain morphisms $i : P'_\bullet \to P_\bullet$ and $\pi : P_\bullet \to P''_\bullet$ so that

$$0 \longrightarrow P'_\bullet \xrightarrow{i} P_\bullet \xrightarrow{\pi} P''_\bullet \longrightarrow 0$$

is an exact sequence of complexes and so that the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & P'_0 \arrow[r, "i_0"] \arrow[d, "\epsilon'"] & P_0 \arrow[r, "\pi_0"] \arrow[d, "\epsilon"] & P''_0 \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d] & M \arrow[r, "\beta"] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

For future reference, we will express this diagram as

$$\begin{tikzcd}
0 \arrow[r] & P'_\bullet \arrow[r, "i"] \arrow[d, "\epsilon'"] & P_\bullet \arrow[r, "\pi"] \arrow[d, "\epsilon"] & P''_\bullet \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d] & M \arrow[r, "\beta"] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Proposition 14. *Any short exact sequence of modules*

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

has a projective resolution.

Proof. We construct these chain morphisms inductively. Begin with projective presentations for each of M' and M''.

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & N_0' \arrow[d] & & N_0'' \arrow[d] & \\
 & P_0' \arrow[d] & & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & & 0 &
\end{tikzcd}$$

Let $P_0 = P_0' \oplus P_0''$. This is projective to get

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & N_0' \arrow[d] & & N_0'' \arrow[d] & \\
 & P_0' \arrow[r, "i_0"] \arrow[d] & P_0 \arrow[r, "\pi_0"] & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & & 0 &
\end{tikzcd}$$

with i_0, π_0 forming a split exact sequence. To get the augmentation map $\epsilon : P_0 \to M$ start by lifting $\epsilon'' : P_0'' \to M''$ to $\epsilon^* : P_0'' \to M$. Thus $\beta \circ \epsilon^* = \epsilon''$. Define $\epsilon : P_0 \to M$ by $\epsilon(x_0', x_0'') = \alpha\epsilon'(x_0') + \epsilon^*(x_0'')$. This diagram clearly commutes

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & N_0' \arrow[d] & & N_0'' \arrow[d] & \\
 & P_0' \arrow[r, "i_0"] \arrow[d] & P_0 \arrow[r, "\pi_0"] & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\epsilon'"] & M \arrow[r, "\beta"] \arrow[d, "\epsilon"] & M'' \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Let $N_0 = \ker \epsilon$. So we get

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
 & N_0' \arrow[r, "i_0"] \arrow[d] & N_0 \arrow[r, "\pi_0"] & N_0'' \arrow[d] & \\
 & P_0' \arrow[r, "i_0"] \arrow[d] & P_0 \arrow[r, "\pi_0"] & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\epsilon'"] & M \arrow[r, "\beta"] \arrow[d, "\epsilon"] & M'' \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

We want surjectivity of ϵ and injectivity $N_0 \to P_0$ exact. We can do the first by the Snake lemma, and the second by either the 5-lemma of the Snake Lemma.

Next, we obtain the square

$$
\begin{tikzcd}
0 \arrow[r] & N_1' \arrow[r, "i_1"] \arrow[d] & N_1 \arrow[r, "\pi_1"] \arrow[d] & N_1'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_1' \arrow[r, "i_1"] \arrow[d] & P_1 \arrow[r, "\pi_1"] \arrow[d] & P_1'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N_0' \arrow[r, "i_0"] \arrow[d] & N_0 \arrow[r, "\pi_0"] \arrow[d] & N_0'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

by the same exact method. Thus we are left with the following chain complexes

$$
\begin{tikzcd}
0 \arrow[r] & P_1' \arrow[r] \arrow[d] \arrow[dd, bend right, "\partial_2'"] & P_1 \arrow[r] \arrow[d] \arrow[dd, bend right, "\partial_2'"] & P_1'' \arrow[r] \arrow[d] \arrow[dd, bend right, "\partial_2'"] & 0 \\
0 \arrow[r] & N_0' \arrow[r] \arrow[d] & N_0 \arrow[r] \arrow[d] & N_0'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_0' \arrow[r] \arrow[d] & P_0 \arrow[r] \arrow[d] & P_0'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

which gives our projective resolutions of

$$
0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0
$$

□

Theorem 8. *Let F be a covariant additive functor from R-<u>mod</u> to $\mathbb{Z}$-<u>mod</u>, and let*

$$
0 \to M'' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \to 0
$$

Then for any $n \geq 1$ there is a conntecting homomorphism $\Delta_n : L_nF(M'') \to L_{n-1}F(M')$ such that we obtain a long exact sequence of left derived functors

$$\cdots \to L_n(F(M')) \to L_n(F(M)) \to L_n(F(M'')) \to L_{n-1}(F(M')) \to \cdots$$

Proof. We begin with projective resolution

$$\begin{tikzcd}
0 \arrow[r] & P'_\bullet \arrow[r, "i"] \arrow[d, "\epsilon'"] & P_\bullet \arrow[r, "\pi"] \arrow[d, "\epsilon"] & P''_\bullet \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d] & M \arrow[r, "\beta"] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

For each $n \geq 0$ we have a short exact sequence

$$0 \longrightarrow P'_n \longrightarrow P_n \longrightarrow P''_n \longrightarrow 0$$

As P''_n is projective, we can view $P_n \cong P'_n \oplus P''_n$. Now applying F we see $F(P_n) \cong F(P'_n) \oplus F(P''_n)$ by the additivity of F. Therefore the sequence

$$0 \longrightarrow F(P'_n) \longrightarrow F(P_n) \longrightarrow F(P''_n) \longrightarrow 0$$

remains short exact.

$$\begin{tikzcd}
0 \arrow[r] & F(P'_\bullet) \arrow[r, "F(i)"] \arrow[d, "F(\epsilon)'"] & F(P_\bullet) \arrow[r, "F(\pi)"] \arrow[d, "F(\epsilon)"] & F(P''_\bullet) \arrow[r] \arrow[d, "F(\epsilon'')"] & 0 \\
0 \arrow[r] & F(M') \arrow[r, "F(\alpha)"] \arrow[d] & M \arrow[r, "F(\beta)"] \arrow[d] & F(M'') \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

is still a short exact sequence of complexes. Applying the homology functors, we get a connecting homomorphism $\delta_n : H_n(P''_\bullet) \to H_{n-1}(P'_\bullet)$ creating the long exact sequence of homology

$$\cdots \longrightarrow H_n(F(P'_\bullet)) \longrightarrow H_n(F(P_\bullet)) \longrightarrow H_n(F(P''_\bullet)) \longrightarrow H_{n-1}(F(P'_\bullet)) \longrightarrow \cdots$$

and this by definition is

$$\cdots \longrightarrow L_n(F(M')) \longrightarrow L_n(F(M)) \longrightarrow L_n(F(M'')) \longrightarrow L_{n-1}(F(M')) \longrightarrow \cdots$$

□

3.12.1 Right Derived Functors for Covariant Functors

Let $F : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ be an additive functor. The right derived functors come from cohomology.

Take a module M and an injective resolution $(Q^\bullet, \eta)$ of M

$$0 \longrightarrow M \xrightarrow{\eta} Q^0 \xrightarrow{d^0} Q^1 \longrightarrow \cdots$$

with each Q^i injective. Then applying F we get

$$0 \longrightarrow F(M) \xrightarrow{F(\eta)} F(Q^0) \xrightarrow{F(d^0)} F(Q^1) \longrightarrow \cdots$$

Considering the sequence

$$0 \longrightarrow F(Q^0) \xrightarrow{F(d^0)} F(Q^1) \xrightarrow{F(d^1)} F(Q^2) \longrightarrow \cdots$$

and $R^nF(M) := H^n(F(Q^\bullet))$. In particular, $R^0F(M) = \ker F(d^0)$.

If $\lambda \in \hom_R(M', M)$, then we can lift λ to $\Lambda : (Q^{\bullet'}, \eta') \to (Q^\bullet, \eta)$ injective resoltions of M' and M respectively. Then $R^nF(\lambda) = \widetilde{F(\Lambda^n)} : R^nF(M') \to R^nF(M)$.

Definition 37. R^nF so defined are the *right derived functors* for F. They are covariant and additive.

There is also a long exact sequence of right derived functors

$$\cdots \to R^n(F(M')) \to R^nF(M) \to R^nF(M'') \to R^{n+1}F(M') \to R^{n+1}F(M) \to \cdots$$

associated to a short exact sequence

$$0 \to M' \to M \to M'' \to 0$$

We get this by taking compatible injective resolutions

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\eta'"] & M \arrow[r, "\beta"] \arrow[d, "\eta"] & M'' \arrow[r] \arrow[d, "\eta''"] & 0 \\
0 \arrow[r] & Q^{\bullet'} \arrow[r] & Q^\bullet \arrow[r] & Q^{\bullet''} \arrow[r] & 0
\end{tikzcd}$$

and use the Snake Lemma to get our connecting homomorphisms; as the right derived functors are cohomology functors, this induces maps on the chomology groups defining R^nF.

3.12.2 Derived Functors for Contravariant Functors

> "This is just like eating crawfish... You've ever eaten crawfish? You pull off the heads"
>
> Cogdell

Let $G : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ contravariant and additive functors. We will now construct the left and right derived functors.

Left derived functors still correspond to homology, and the right derived functors correspond to cohomology.

For left derived functors, begin with a module M and an injective resolution $(Q^\bullet, \eta)$

$$0 \to M \to Q^0 \to Q^1 \to Q^2 \to \cdots$$

applying G yields

$$\cdots \to G(Q^2) \to G(Q^1) \to G(Q^0) \to G(M) \to 0$$

considering the complex

$$\cdots \to G(Q^2) \to G(Q^1) \to G(Q^0) \to 0$$

and taking homology, we define $L_nG(M) = H_n(G(Q^\bullet)$. Thus $L_0G(M) = \operatorname{coker} G(d^0)$.

On a morphism $\lambda : M' \to M$ we lift λ to $\Lambda : (Q^{\bullet'}, \eta') \to (Q^\bullet, \eta)$. Applying G and taking homology we get maps on homology $L_nG(\lambda) : L_nG(M) \to L_nG(M')$.

Definition 38. L_nG so defined are the left derived functors of G. They are contravariant and additive.

There are connecting homomorphisms

$$\Delta_n : L_nG(M') \to L_{n-1}G(M'')$$

We get these by taking an exact sequence of R-modules

$$0 \to M' \to M \to M'' \to 0$$

then taking injective resolutions

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\eta'"] & M \arrow[r, "\beta"] \arrow[d, "\eta"] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Q^{\bullet'} \arrow[r] & Q^\bullet \arrow[r] & Q^{\bullet''} \arrow[r] & 0
\end{tikzcd}$$

and apply G and take homology. With these connecting homomorphisms we get the following long exact sequence

$$\cdots \to L_nG(M'') \to L_nG(M) \to L_nG(M') \to L_{n-1}G(M'') \to L_{n-1}G(M) \to \cdots$$

Similarly for the right derived functors of G we begin with M and a projective resolution $(P_\bullet, \epsilon)$

$$\cdots \to P_2 \to P_1 \to P_0 \to M \to 0$$

apply G to get

$$0 \to G(M) \to G(P_0) \to G(P_1) \to G(P_2) \to \cdots$$

reduce to

$$0 \to G(P_0) \to G(P_1) \to G(P_2) \to \cdots$$

taking cohomology

$$R^nG(M) = H^n(G(P_\bullet))$$

and $R^0G(M) = \ker G(d^1)$.

If $\mu : M' \to M$ is a morphism, we lift μ to a morphism of resolution $(P'_\bullet, \epsilon') \to (P_\bullet, \epsilon)$ then apply G, then take out $G(M), G(M')$ in the sequences and get the induced maps on cohomology. This gives up maps $R^nG(\mu) : R^nG(M) \to R^nG(M')$.

Definition 39. The $R^nG(M)$ so defined are the *right derived functors* for G. They are contravariant, additive functors.

In a similar way we get a long exact sequence of right derived functors

$$R^nG(M'') \to R^nG(M) \to R^nG(M') \to R^{n+1}G(M'') \to R^{n+1}G(M) \to \cdots$$

from a short exact sequence of R-modules

$$0 \to M' \to M \to M'' \to 0$$

3.13 Ext

The functor $\operatorname{Ext}^n_R(-,N)$ is the right derived functor of the contravariant hom functor $\hom_R(-,N)$. Recall that $\hom_R(-,N)$ is contravariant and additive, and if $\alpha \in \hom_R(M,M')$ then $\hom_R(\alpha,N) : \hom_R(M',N) \to \hom_R(M,N)$ by $f \mapsto f \circ \alpha$. Also, the functor $\hom_R(-,N)$ is left-exact.

Define $\operatorname{Ext}^n_R(-,N) = R^n \hom_R(-,N)$.

Proposition 15. *1.* $\operatorname{Ext}^0_R(M,N) \cong \hom(M,N)$

2. Ext^1_R *measures the failure of exactness of* $\hom(-,N)$.

3. If

$$0 \to K \xrightarrow{\eta} P \xrightarrow{\epsilon} M \to 0$$

is a projective presentation of M *then* $\operatorname{Ext}^1(M,N) = \operatorname{coker} \eta^* = \hom(K,N)/\eta^*(\hom(P,N))$.

Proof. 1. To compute $\operatorname{Ext}^0_R(-,N)$, begin with a projective resolution of M

$$\cdots \to P_1 \to P_0 \to M \to 0$$

Then the right derived functors are given by the cohomology of

$$0 \to \hom_R(P_0,N) \xrightarrow{d_1^*} \hom_R(P_1,N) \to \hom_R(P_2,N) \to \cdots$$

Thus $\operatorname{Ext}^0_R(M,N) = \ker d_1^*$. As $\hom(-,N)$ is left exact, we have the exact sequence

$$0 \to \hom(M,N) \xrightarrow{\epsilon^*} \hom(P_0,N) \xrightarrow{d_1^*} \hom(P_1,N)$$

so $\ker d_1^* = \epsilon^*(\hom(M,N)) \cong \hom(M,N)$. This is also a natural isomorphism by Exercise 33.

2. Start with a short exact sequence

$$0 \to M' \to M \to M'' \to 0$$

Left exactness gives

$$0 \to \hom(M'',N) \to \hom(M,N) \to \hom(M',N) \to 0$$

exact. By the long exact sequence of right derived functors

$$0 \to \operatorname{Ext}^0(M'',N) \to \operatorname{Ext}^0(M,N) \to \operatorname{Ext}^0(M',N) \to \operatorname{Ext}^1(M'',N) \to \cdots$$

replacing Ext^0 with hom we then get

$$0 \to \hom(M'',N) \to \hom(M,N) \to \hom(M',N) \to \operatorname{Ext}^1(M'',N) \to \cdots$$

Thus the map $\hom(M,N) \to \hom(M',N)$ is surjective if and only if $\operatorname{Ext}^1(M'',N) = 0$.

3. We have the exact sequence

$$0 \to \hom(M,N) \to \hom(P,N) \to \hom(K,N) \to \operatorname{Ext}^1(M,N) \to \operatorname{Ext}^1(P,N) = 0$$

□

Theorem 9 (Characterization of Projective Modules with Ext)**.** *The following conditions are equivalent:*

1. *M is projective.*
2. $\mathrm{Ext}^1(M, N) = 0$ *for all* $n \geq 1$ *and all* N*.*
3. $\mathrm{Ext}^1(M, N) = 0$ *for all* N*.*

Proof. $1 \Rightarrow 2$) If M is projective, we have the projective resolution

$$\cdots \to 0 \to 0 \to M \to M \to 0$$

via id_M. Thus $\mathrm{Ext}^n(M, N) = 0$ for $n \geq 1$.

$2 \Rightarrow 3$) Trivial

$3 \Rightarrow 1$) Begin with a projective presentation of M

$$0 \to K \to P \to M \to 0$$

with P projective. Applying the long exact sequence of derived functors yields

$$0 \to \hom(M, N) \to \hom(P, N) \to \hom(K, N) \to \mathrm{Ext}^1(M, N) = 0 \to \mathrm{Ext}^1(P, N) = 0$$

for all N. Now take $N = K$. So

$$0 \to \hom(M, K) \to \hom(P, K) \to \hom(K, K) \to 0$$

is exact. Thus there is some $\phi \in \hom(P, K)$ that maps to id_K. This map ϕ then splits the exact sequence

$$0 \to K \to P \to M \to 0$$

implying M is a direct summand of P, a projective module, so M is projective. □

3.13.1 The Meanining of Ext^1

"Legally that's not disallowed"

Cogdell

Before the Ext functor came about, there was Baer's theory of extensions of modules:

Definition 40. If M and N are R-modules, an R-module E is called an *extension of M by N* if there is a short exact sequence

$$0 \to N \to E \to M \to 0$$

In the case E_1 and E_2 are extensions of M by N, then E_1 and E_2 are *equivalent* if there is $\gamma \in \hom(E_1, E_2)$ so that

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r] \arrow[d] & E_1 \arrow[r] \arrow[d, "\gamma"] & M \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N \arrow[r] & E_2 \arrow[r] & M \arrow[r] & 0
\end{tikzcd}$$

By the 5-Lemma, this is an isomorphism.

Note that extensions always exist, as we can take $E = M \oplus N$. Then we can consider $E(M, N)$, the equivalence classes of extensions of M by N.

Theorem 10. *$E(M, N) \cong \mathrm{Ext}^1(M, N)$ as sets*

Moreover, $E(M, N)$ has the structure of a group (defined by Baer), and this map is an isomorphism of abelian groups.

Consider $\mathrm{Ext}^n(M, N)$ as a functor in the second variable. On objects, we have $N \mapsto \mathrm{Ext}^n(M, N)$. On morphisms, if $\beta \in \hom(N, N')$ then

$$\begin{tikzcd}
0 \arrow[r] & \hom(M,N) \arrow[r] \arrow[d, "\beta_*"] & \hom(P_0,N) \arrow[r] \arrow[d, "\beta_*^0"] & \hom(P_1,N) \arrow[r] \arrow[d, "\beta_*^1"] & \cdots \\
0 \arrow[r] & \hom(M,N') \arrow[r] & \hom(P_0,N') \arrow[r] & \hom(P_1,N') \arrow[r] & \cdots
\end{tikzcd}$$

and these maps induce maps $\widetilde{\beta_*^n}$ on cohomology. So

$$\widetilde{\beta_*^n} : \mathrm{Ext}^n(M, N) \to \mathrm{Ext}^n(M, N')$$

Defining a functor with these maps, we have $\mathrm{Ext}^n(M, -)$ is covariant and additive.

If we start with an exact sequence

$$0 \to N' \xrightarrow{\alpha} N \xrightarrow{\beta} N'' \to 0$$

and we take a projective resolution $(P_\bullet, \epsilon)$ of M we get maps

$$\hom(P_\bullet, N') \to \hom(P_\bullet, N) \to \hom(P_\bullet, N'')$$

and at each stage,

$$\hom(P_n, N') \xrightarrow{\alpha_*^n} \hom(P_n, N) \xrightarrow{\beta_*^n} \hom(P_n, N'')$$

As P_n is projective, $\hom(P_n, -)$ is exact. Therefore for all n we have the short exact sequence

$$0 \to \hom(P_n, N') \xrightarrow{\alpha_*^n} \hom(P_n, N) \xrightarrow{\beta_*^n} \hom(P_n, N'') \to 0$$

so we have the short exact sequence of complexes

$$0 \to \hom(P_\bullet, N') \to \hom(P_\bullet, N) \to \hom(P_\bullet, N'') \to 0$$

and this gives a long exact sequence of cohomology. As each $H^n(\hom(P_\bullet, N)) = \mathrm{Ext}^n(M, N')$, this actually gives us a long exact sequence of $\mathrm{Ext}(M, -)$.

This long exact sequence allows us to characterize injective modules.

Theorem 11. *The following are equivalent:*

1. *N is injective.*
2. *$\mathrm{Ext}^n(M, N) = 0$ for all $n \geq 1$ and all M.*
3. *$\mathrm{Ext}^1(M, N) = 0$ for all M.*

Proof. $1 \Rightarrow 2)$ N is injective if and only if $\hom(-, N)$ is exact. Thus if we take a projective resolution $(P_\bullet, \epsilon)$, then we get the exact sequence

$$0 \to \hom(M, N) \to \hom(P_0, N) \to \hom(P_1, N)$$

is exact, whence $\mathrm{Ext}^0(M, N) = \hom(M, N)$ and $\mathrm{Ext}^n(M, N) = 0$ for $n \geq 1$ by exactness. $2 \Rightarrow 3)$ Trivial. $3 \Rightarrow 1)$ We assume $\mathrm{Ext}^1(M, N) = 0$ for all M, and take an injective presentation of N

$$0 \to N \to Q \to L \to 0$$

with Q injective. Applying the long exact sequence of Ext we get

$$0 \to \hom(M, N) \to \hom(M, Q) \to \hom(M, L) \to 0 = \mathrm{Ext}^1(M, N)$$

for all M. In particular,

$$0 \to \hom(L, N) \to \hom(L, Q) \to \hom(L, L) \to 0$$

and there exists $\lambda \in \hom(L, Q)$ with $\lambda \mapsto \mathrm{id}_L$ which says the exact sequence

$$0 \to N \to Q \to L \to 0$$

splits, and N is a direct summand of the injective module Q and therefore N is injective. $\square$

Another approach to Ext is to start with $F = \hom(M, -)$ the covariant hom functor and then defined $\mathrm{Ext}^n(M, -) = R^n F$, we will call this functor $\overline{\mathrm{Ext}}^n(M, -)$. This would be computed by taking an injective resolution of N

$$0 \to N \to Q^0 \to Q^1 \to \cdots$$

and taking cohomology after applying $\hom(M, -)$. Thus $\overline{\mathrm{Ext}}^0(M, N) \cong \hom(M, N)$ and $\overline{\mathrm{Ext}}^n(M, N) = H^n(\hom(M, Q^\bullet))$. This definition of $\overline{\mathrm{Ext}}$ allows us to develop the same tools as with Ext.

Theorem 12. $\overline{\mathrm{Ext}}^n$ *and* Ext^n *are isomorphic for bifunctors.*

Theorem 13. *For all* M, N, $\mathrm{Ext}^n(M, N) \cong \overline{\mathrm{Ext}}^n(M, N)$

Proof. Induct on n. The case $n = 0$ follows from $\mathrm{Ext}^0(M, N) = \hom(M, N) = \overline{\mathrm{Ext}}^0(M, N)$. For $n = 1$ take a projective presentation of M

$$0 \to K \to P \to M \to 0$$

Applying the long exact sequence of Ext or $\overline{\mathrm{Ext}}$ in the first variable

$$0 \to \hom(M, N) \to \hom(P, N) \to \hom(K, N) \to \mathrm{Ext}^1(M, N) \to \mathrm{Ext}^1(P, N) = 0$$

$$0 \to \hom(M, N) \to \hom(P, N) \to \hom(K, N) \to \overline{\mathrm{Ext}}^1(M, N) \to \overline{\mathrm{Ext}}^1(P, N) = 0$$

thus $\mathrm{Ext}^1(M, N) \cong \hom(K, N)/\hom(P, N) \cong \overline{\mathrm{Ext}}^1(M, N)$. The inductive step then acts in the same way. We get

$$\mathrm{Ext}^n(M, N) \cong \mathrm{Ext}^{n-1}(K, N) \cong \overline{\mathrm{Ext}}^{n-1}(K, N) \cong \overline{\mathrm{Ext}}^n(M, N)$$

by the inductive hypothesis. $\square$

3.13.2 Adjointness of Ext and hom

Let R and S be objects in $\underline{Ring}$ and A left R-module, C left S-module, and B a left S-module and right R-module.

Then adjointness says $\hom_S(B \otimes_R A, C) \cong \hom_R(A, \hom_S(B, C))$. We can check $B \otimes_R A$ is a left S-module, and $\hom_S(B, C)$ is a left R-module. The isomorphism is given by $f \mapsto g_a$ where $g_a(b) = f(b \otimes a)$.

The inverse map is given by $g \mapsto g'$ with $g'(b \otimes a) = g(a)(b)$.

3.14 Tor

Tor is given by the derived functors for $\otimes$. As $\otimes$ is right exact, we use the left derived functors.

Consider $M \otimes -$ for a right R-module M. This is a functor from R-$\underline{\text{mod}}$ and $\mathbb{Z}$-$\underline{\text{mod}}$, and it is covariant, additive, and right exact. On morphisms, $\alpha : N \to N'$ yields a map $\mathrm{id}_M \otimes \alpha : M \otimes N \to M \otimes N'$ given on pure tensors by $\mathrm{id}_M \otimes \alpha(m \otimes n) = m \otimes \alpha(n)$.

Definition 41. $\mathrm{Tor}_n(M, -) = L_n(M \otimes -)$

To compute this, we begin with a projective resolution of N

$$\cdots \to P_2 \to P_1 \to P_0 \xrightarrow{\epsilon} N \to 0$$

and then take the homology of the sequence

$$\cdots M \otimes P_2 \to M \otimes P_1 \to M \otimes P_0 \to 0$$

so $\mathrm{Tor}_n(M, N) = H_n(M \otimes P_\bullet)$. We also recover $M \otimes -$ by

$$\mathrm{Tor}_0(M, N) = \operatorname{coker} \mathrm{id}_M \otimes d_1 = M \otimes P_0 / \operatorname{im}(\mathrm{id}_M \otimes d_1) = M \otimes N$$

Proposition 16. *1.* $\mathrm{Tor}_0(M, N) \cong M \otimes N$

2. Tor_1 *measures failure of exactness of* $M \otimes -$

Proof. 1. This was shown above.

2. If we have a short exact sequence

$$0 \to N' \to N \to N'' \to 0$$

right exactness gives

$$M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$$

and the long exact sequence of derived functors gives

$$\cdots \to \mathrm{Tor}_1(M, N) \to \mathrm{Tor}_1(M, N'') \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$$

Thus in the case M is flat, we have $\mathrm{Tor}_1(M, N'') = 0$.

□

Just as Ext gave us a characterization of projectives and injectives, Tor gives an equivalent definition for flatness.

Theorem 14. *The following are equivalent:*

1. *M is flat.*

2. $\mathrm{Tor}_n(M,N) = 0$ *for* $n \geq 1$ *and all* N.

3. $\mathrm{Tor}_1(M,N) = 0$ *for all* N .

Proof. $1 \Rightarrow 2$) In the case M is flat, $M \otimes -$ is exact, whence if $(P_\bullet, \epsilon)$ is a projective resolution, then

$$\cdots \to M \otimes P_2 \to M \otimes P_1 \to M \otimes P_0 \to 0$$

is exact everywhere except at $M \otimes P_0$. Thus the n-th homology is trivial for $n \geq 1$ and $\mathrm{Tor}_n(M,N) = 0$ for $n \geq 1$.

$2 \Rightarrow 3$) Trivial

$3 \Rightarrow 1$) Consider the long exact sequence of homology of the short exact sequence

$$0 \to N' \to N \to N'' \to 0$$

$$\cdots \to \mathrm{Tor}_1(M,N'') = 0 \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$$

and so $M \otimes -$ is exact, and thus M is flat. □

Now we consider $\mathrm{Tor}_n(M,N)$ as a functor in the first variable. In the case $\alpha \in \hom(M,M')$ and $(P_\bullet, \epsilon)$ is a projective resolution of N, α induces maps $\alpha \otimes \mathrm{id}_{P_i}$ on the resolutions

$$\begin{tikzcd}
\cdots \arrow[r] & M \otimes P_1 \arrow[r] \arrow[d] & M \otimes P_0 \arrow[r] \arrow[d] & M \otimes N \arrow[r] \arrow[d, "\alpha \otimes id_N"] & 0 \\
\cdot \arrow[r] & M' \otimes P_1 \arrow[r] & M' \otimes P_0 \arrow[r] & M' \otimes N \arrow[r] & 0
\end{tikzcd}$$

Thus we have a map $\widetilde{\alpha \otimes P_n} : H_n(M \otimes P_\bullet) \to H_n(M' \otimes P_\bullet)$, and this we define to be $\mathrm{Tor}_n(\alpha, N)$.

This makes $\mathrm{Tor}_n(-,N)$ a covariant and additive functor.

Given an exact sequence

$$0 \to M' \to M \to M'' \to 0$$

take a projective resolution $(P_\bullet, \epsilon)$ of N to get exact sequences

$$0 \to M' \otimes P_n \to M \otimes P_n \to M'' \otimes P_n \to 0$$

This gives us a short exact sequence of complexes

$$0 \to M' \otimes P_\bullet \to M \otimes P_\bullet \to M'' \otimes P_\bullet \to 0$$

so we get a long exact sequence of homology and thus Tor

$$\cdots \to \mathrm{Tor}_1(M,N) \to \mathrm{Tor}_1(M'',N) \to \mathrm{Tor}_0(M',N) \to \mathrm{Tor}_0(M,N) \to \mathrm{Tor}_0(M'',N) \to 0$$

We could also have developed Tor as the left derived functors of $- \otimes N$; call this construction $\overline{\mathrm{Tor}_n}(-,N) = L_n(- \otimes N)$. Similarly, $\overline{\mathrm{Tor}_0}(M,N) = M \otimes N$.

Theorem 15. *For* $M \in \mathrm{ob}(\underline{mod} - R)$ *and* $N \in \mathrm{ob}(R - \underline{mod})$, $\mathrm{Tor}_n(M,N) \cong \overline{\mathrm{Tor}_n}(M,N)$.

The proof uses induction using the long exact sequences, just as the case for Ext and $\overline{\mathrm{Ext}}$.

From this we have the characterization of flatness for $R - \underline{mod}$.

Theorem 16. *The following are equivalent:*

1. *N is flat as an R-module.*

2. $\mathrm{Tor}_n(M, N) = 0$ *for* $n \geq 1$ *and all* M.

3. $\mathrm{Tor}_1(M, N) = 0$ *for all* M.

Proof. The proof is exactly the same as the argument in the first variable. □

We will see that Tor_1 relates in some way to the *torsion* of a module.

4 Fields and Galois Theory

At this point in the semester, coronavirus has hit the U.S. and schools have been moved online. Thus, these notes are just copied from the ones posted online, and there will be no more quotes.

Definition 42. If E is a field, and $F \subset E$ is also a field (inheriting the same operations as in E), we say F is a *subfield* of E and E is an *extension field* of F.

In the case E/F is an extension, we can view E as a vector space over F by forgetting the multiplication of E on itself. In this way, there is an idea of the degree of the dimension, as the degree of the vector space.

Definition 43. The *degree* of an extension E/F is denoted $(E : F) = \dim_F(E)$.

Note that extensions are transitive, i.e., if $F \subset E \subset K$, then $F \subset K$. In this case, the degree of the extensions is multiplicative.

Theorem 17. *Let $F \subset E \subset K$ be fields. Then $(K : F)$ is finite if and onliy if $(K : E)$ and $(E : F)$ are finite, in which case*

$$(K : F) = (K : E)(E : F)$$

Proof. In the case $(K : E)$ and $(E : F)$ are finite, we have $\{\alpha_1, \ldots, \alpha_d\}$ a basis of K over E and $\{\beta_1, \ldots, \beta_t\}$ a basis of E over F. The claim is that $\{\alpha_i\beta_j\}_{1\leq i\leq d,\, 1\leq j\leq t}$ is a basis of K over F. This is easily checked to be the case and gives the proper multiplication of degrees.

Now suppose K/F is finite. Then $(E : F) \leq (K : F)$, as any linear independent subset of E over F remains linearlly independent in K. Similarly, $(K : E) \leq (K : F)$ as any spanning set of K over F still spans K over E. □

We get the following corollary immediately through induction:

Corollary 2. *If $F \subset F_1 \subset \cdots \subset F_n$ is a chain of field extensions, then*

$$(F_n : F) = (F_n : F_{n-1})(F_{n-1} : F_{n-2}) \cdots (F_1 : F)$$

If we have a field extension E/F, we can create intermediate rings and fields by appending elements of E to F.

Definition 44. If E/F is a field extension, and $S \subset E$, define $F[S]$ to be the subring of E generated by F and S, and define $F(S)$ to be the subfield of E generated by F and S.

In the case S is finite, we can view these rings and fields as the images of polynomial rings and fields of rational functions under the evaluation homomorphism. More specifically, if $S = \{\alpha_1, \ldots, \alpha_n\}$ then $F[S]$ is the image of

$$F[x_1, \ldots, x_n] \to E$$

where

$$x_i \mapsto \alpha_i$$

and $F(S)$ is the image of

$$F(x_1, \ldots, x_n) \to E$$

where

$$x_i \mapsto \alpha_i$$

4.1 Algebraic Elements

Definition 45. Let E/F be a field extension, and $\alpha \in E$. We say α is *algebraic* over F if it is the root of a non-zero polynomial in $F[x]$.

For any $\alpha \in E$ we have the map $F[x] \to E$ by $f(x) \mapsto f(\alpha)$. Then α is algebraic if and only if this map has a non-trivial kernel, I. As $F[x]$ is a PID, if I is non-trivial, I is generated by a (non-zero) polynomial $f(x)$ that is monic and irreducible.

Definition 46. We call this generating polynomial $f(x)$ the *minimal polynomial* of α.

Theorem 18. *Let E/F be an extension of fields. Then $\alpha \in E$ is algebraic over F if and only if $(F(\alpha) : F) < \infty$.*

Proof. In the case α is algebraic, its minimal polynomial $f_\alpha(x) = x^n + a_1 x^{n-1} + \cdots a_n$ is irreducible, so $F[\alpha] = F(\alpha)$ is a field (by the above, $F[\alpha] = F[x]/(f_\alpha(x))$). Via the division algorithm, we see that $F[\alpha] \cong F[x]/(f_\alpha(x))$ is spanned by $x^0, \ldots, x^{n-1}$. Thus $(F(\alpha) : F) = n < \infty$.

In the case α is not algebraic, any solution of a linear equation involving only $\alpha^0, \alpha^1, \ldots$ must be the trival solution or else α would be the root of some non-zero polynomial over F. This shows $F(\alpha) : F$ has an infinite set of linearly idependent elements, whence $(F(\alpha) : F) = \infty$. □

While it is not the case that for arbitrary $f(x) \in F[x]$ we have a root in F, a theorem by Kronecker guarantees the existence of an extension E/F in which $f(x)$ has a root.

Theorem 19 (Kronecker)**.** *If $f(x) \in F[x]$ is a polynomial with $\deg f \geq 1$ then there is an extension E/F in which $f(x)$ has a root.*

Proof. By factoring f into irreducibles, we may assume f itself is monic and irreducible—thus

$$f(x) = x^n + a_1 x^{n-1} + \cdots + a_n$$

In the case $n = 1$, then $-a_n$ is a root of $f(x)$ in F. Otherwise consider $E = F[x]/(f(x))$. As $f(x)$ is irreducible, this is a field, and contains F. The element $x + (f(x))$ is then a root of $f(x)$. □

By the construction we also have a loose bound on the degree of the extension, namely

$$(E : F) = \deg f'$$

were f' is an irreducible factor of the polynomial f in question. This sort of extension is also unique up to isomorphism, as the next result shows.

Theorem 20. *Let $\sigma : F \to F'$ be an isomorphism of fields. Let $f(x) \in F[x]$ be irreducible, $f'(x) = \sigma(f(x)) \in F'[x]$ (this is irreducible). If $E = F(\beta)$ where $f(\beta) = 0$ and $E' = F'(\beta')$ where $f'(\beta') = 0$ then σ can be extended to an isomorphism $\sigma : E \to E'$.*

Proof. Note that $F(\beta) \cong F[x]/(f(x))$ and $F'(\beta') \cong F'[x](f'(x))$ and the map σ induces an isomorphism between these latter representations. □

4.2 Splitting Fields

We have shown the existence of fields in which a polynomial has all of its roots. We now consider the minimal such field.

Definition 47. Let F be a field, $f(x) \in F[x]$ monic. An extension E/F is called a *splitting field* of $f(x)$ over F if

$$f(x) = (x - \alpha_1) \cdots (x - \alpha_n)$$

and

$$E = F(\alpha_1, \ldots, \alpha_n)$$

Note that as E is obtained by adjoining finitely many algebraic elements, $(E : F) < \infty$.

Theorem 21. *If $p(x) \in F[x]$ is a monic polynomial $\deg p \geq 1$, then there exists a splitting field E of $p(x)$.*

Proof. Begin by factoring $p(x)$ into irreducibles in $F[x]$

$$p(x) = f_1(x) \cdots f_r(x)$$

We now induct on the order of $n - r$. In the case $n - r = 0$, then $p(x)$ already splits in $F[x]$ and we are done. In the case $n - r > 0$, then without loss of generality $\deg f_1 > 1$, and we can find an extension K/F for which f_1 has a root α_1 and $K = F(\alpha_1)$, and so $f_1(x) = (x - \alpha_1)g_1(x)$. Factoring $p(x)$ in $K[x]$ then decreases $n - r$ by one, so by induction there is a field E/K for which p splits, and it will be given by $E = K(\alpha_1, \ldots, \alpha_n) = F(\alpha_1)(\alpha_1, \ldots, \alpha_n) = F(\alpha_1, \ldots, \alpha_n)$. □

Just as extensions of a field to give a root of a polynomial are unique up to isomorphism, so too are splitting fields.

Theorem 22 (Uniqueness of Splitting Fields)**.** *Let $\sigma : F \to F'$ be an isomorphism of fields. Let $p(x)$ in$F[x]$ and $p'(x) = \sigma(p(x)) \in F'[x]$. Let E/F be a splitting field of $p(x)$ and E'/F' a splitting field of $p'(x)$. Then σ can be extended to an isomorphism $\sigma' : E' \to F'$.*

Proof. If $f(x) \in F[x]$ is an irreducible factor of p, then E contains a root of $f(x)$. If all the roots of $p(x)$ lie in F, then it is easy to see all roots of p' lie in F', and then σ is the isomorphism of the splitting fields.

Now we induct on the number of roots of p not in F. Begin by factoring p into irreducible factors $p(x) = f_1(x) \cdots f_m(x)$. Not all these factors have degree 1, so without loss of generality $\deg f_1 > 1$. By applying σ to this factorization, we have $p'(x) = f_1'(x) \cdots f_m'(x)$ where $\sigma(f_i(x)) = f_i'(x)$. Say $f(x)$ has a root $\alpha \in E$ not in F. By the uniqueness of appending a root, we have σ extending to an isomorphism $\sigma_1 : F(\alpha) \to F'(\alpha')$. Now we have E a splitting field of $p(x)$ over $F(\alpha)$ and E' a splitting field of p' over $F'(\alpha')$. However, the number of roots of $p(x)$ outside of $F(\alpha)$ has decreased by 1, so there is an isomorphism $\sigma : E \to E'$ preserving σ_1 and therefore σ. □

As in immediate corrolary by taking σ to be the identity

Corollary 3. *If $p(x) \in F[x]$, then any two splitting fields of $p(x)$ over F are isomorphic.*

4.3 Group Characters

Let G be a mulitplicative group.

Definition 48. Let F be a filed. A homomorphims $\sigma : G \to F$ is called a *character* of G in F.

N.B. Note that this gives a 1-dimensional representation of G with image in F. Specfically, $\sigma(G) \subset F^\times$.

Definition 49. If $\sigma_1, \ldots, \sigma_n : G \to F$ are characters in F, they are called *indepenedent over* F if given coefficents in F such that

$$a_1\sigma_1(x) + \cdots + a_n\sigma_n(x) = 0 \qquad \forall x \in G$$

we have $a_i = 0$ for all i.

As the next result shows, there are only special cases in which characters are dependent.

Theorem 23 (Dedekind's Independence of Characters)**.** *If G is a group and $\sigma_1, \ldots, \sigma_n$ are n* distinct *character of G with values in F, then $\sigma_1, \ldots, \sigma_n$ are independent over F.*

Proof. We induct on n.

In the case $n = 1$, whenever $a_1\sigma_1(x) = 0$ for all x, then $a_1 = a_1\sigma_1(e) = 0$.

For $n > 1$, whenever we have the relation $a_1\sigma_1(x) + \cdots + a_n\sigma_n(x) = 0$ for all $x \in G$ we can assume al $a_i \neq 0$, for as soon as one $a_i = 0$, then we can reduce to a relation containing $n - 1$ distinct characters, and by the inductive hypothesis the relation must then be trivial. Find $\alpha \in G$ for which $\sigma_1(\alpha) \neq \sigma_n(\alpha)$. Multiplying the relatoin by a_n^{-1} we see

$$b_1\sigma_1(x) + \cdots + b_{n-1}\sigma_{n-1}(x) + \sigma_n(x) = 0$$

for all x where $b_i = a_i/a_n$. Then substituting $x \mapsto \alpha x$ we have

$$b_1\sigma_1(\alpha)\sigma_1(x) + \cdots + b_{n-1}\sigma_{n-1}(\alpha)\sigma_{n-1}(x) + \sigma_n(\alpha)\sigma_n(x) = 0$$

so dividing by $\sigma_n(\alpha)$ we have

$$b_1\sigma_1(\alpha)/\sigma_n(\alpha)\sigma_1(x) + \cdots + b_{n-1}\sigma_{n-1}(\alpha)/\sigma_n(\alpha)\sigma_{n-1}(x) + \sigma_n(x) = 0$$

for all $x \in G$. Subtracting these two relations gives

$$[b_1 - b_1\sigma_1(\alpha)/\sigma_n(\alpha)]\,\sigma_1(x) + \cdots + [b_{n-1} - b_{n-1}\sigma_{n-1}(\alpha)/\sigma_n(\alpha)]\,\sigma_{n-1}(x) = 0$$

for all x, whence by the inductive hypothesis $b_1 - b_1\sigma_1(\alpha)/\sigma_n(\alpha) = 0$, but this is a contradiction to the fact $\sigma_1(\alpha) \neq \sigma_n(\alpha$.

Therefore these n characters are independent over F. □

Corollary 4. *If E and E' are two fields and*

$$\sigma_1, \ldots, \sigma_n : E \hookrightarrow E'$$

are distinct field embeddings, then $\sigma_1, \ldots, \sigma_n$ are independnt over E'.

4.4 Fixed Fields

Definition 50. Let $\sigma_1, \ldots, \sigma_n : E \hookrightarrow E'$ be distinct embeddings of fields. An element $\alpha \in E$ is called a *fixed point* of the σ_i if $\sigma_i(\alpha) = \sigma_j(\alpha)$ for all i, j.

It turns out that the set of fixed points forms a subfield of F.

Lemma 4. *Let $\sigma_1, \ldots, \sigma_n : E \hookrightarrow E'$ be distinct field embeddings. Let F be the set of fixed points . Then is a subfield of E.*

Proof. Let $\alpha, \beta \in F$ be fixed points. By the properties of homomorphisms of fields, $\alpha - \beta$, $\alpha\beta^{-1}$ are all fixed points. Thus F is a subfield. □

We can also get a rough bound for the order of the fixed field relative to E.

Theorem 24. *If $\sigma_1, \ldots, \sigma_n : E \hookrightarrow E'$ are distinct and F the fixed field, then $(E : F) \geq n$*

Proof. Suppose for contradiction that $(E : F) = r < n$, and let $w_1, \ldots, w_r$ be a basis of E/F. Then we have the system of homogeneous equations

$$\begin{aligned}
\sigma_1(w_1)x_1 + \sigma_2(w_1)x_2 + \cdots + \sigma_n(w_1)x_n &= 0 \\
\sigma_1(w_2)x_1 + \sigma_2(w_2)x_2 + \cdots + \sigma_n(w_2)x_n &= 0 \\
&\cdots \\
\sigma_1(w_r)x_1 + \sigma_2(w_r)x_2 + \cdots + \sigma_n(w_r)x_n &= 0
\end{aligned}$$

As this system has r equations with n unknowns and $r < n$ there is a non-trivial solution $(\beta_1, \ldots, \beta_n) \in E'$.

By assumption, we can write $y = \sum_{i=1}^{r} a_i w_i$ for $a_i \in F$. For any $y \in F$ we can write $y = \sum a_i w_i$, whence $\sigma_j(y) = \sum \sigma_j(a_i)\sigma_j(w_i) = \sum \sigma_1(a_i)\sigma_j(w_i)$.

Thus

$$\begin{aligned}
\beta_1\sigma_1(y) + \cdots + \beta_n\sigma_n(y) &= \sum \beta_j \sum \sigma_1(a_1)\sigma_j(w_i) \\
&= \sum \sigma_1(a_i) \sum \beta_j\sigma_j(w_i) \\
&= 0
\end{aligned}$$

However, this contradicts Dedekind's theorem of the independence of characters. □

Corollary 5. *Let $\sigma_1, \ldots, \sigma_n \in \mathrm{Aut}(E)$ with $\sigma_1 = \mathrm{id}_E$. If F is the fixed field of these characters, $(E : F) \geq n$.*

4.5 Galois Extensions

Let $F \subset E$ be a subfield of E, and let $\sigma \in \mathrm{Aut}(E)$. We say σ *leaves F fixed* if $\sigma(a) = a$ for all $a \in F$. Note than that the set of automorphisms fixing F then forms a group.

Lemma 5. *If E is an extension of F then the set of automorphisms of E that leave F fixed is a group G, denoted* $\mathrm{Gal}(E/F)$.

N.B. Note that $\mathrm{Gal}(E/F)$ my fix more than just F

Definition 51. If $G \subset \mathrm{Aut}(E)$ define $E^G = \{\alpha \in E : \sigma(\alpha) = \alpha \quad \forall \alpha \in G\}$.

In this case, we have $F \subset E^{\mathrm{Gal}(E/F)}$.

In the case equality holds, we call the extension *Galois*

Definition 52. A (finite) extension E/F is called *Galois* if

- $(E : F) < \infty$
- If $G = \mathrm{Gal}(E/F)$ then $F = E^G$

N.B. Note that Artin calls these sorts of extension *normal*. However, we reserved this word for the following:

Definition 53. An extension E/F is *normal* if it is algebraic and every irreducible $f(x) \in F[x]$ having a root in E splits completely in E. In other words, for every $\alpha \in E$, E contains a splitting field of the minimal polynomial $f_\alpha(x) \in F[x]$.

When we have a Galois extension, we have an exact notion of the degree of extension.

Theorem 25. *Let E be a field and let $G = \{\sigma_1, \ldots, \sigma_n\} \leq \mathrm{Aut}(E)$. If $F = E^G$ then $(E : F) = n$.*

Proof. We have the lower bound $(E : F) \geq n$ by a previous corollary. Now suppose for contradiction that $(E : F) > n$. We then have $\alpha_1, \ldots, \alpha_{n+1} \in E$ that are linearly independent over F. Consider the set of n equations of $n + 1$ unknowns

$$\sum_{i=1}^{n+1} \sigma_j(\alpha_j) x_j =$$

for $j = 1, \ldots, n$. There then exists a non-trivial solution.

Note that this non-trivial solution cannot live in F^{n+1} or else the first equation would read

$$\sum_{i=1}^{n+1} \alpha_i x_i = 0$$

contradicting α_i linearly independent.

We can choose a non-trivial solution containing the least number of non-zero entries, and we can assume the solution takes the form $(\beta_1, \ldots, \beta_r, 0, \ldots, 0)$ with $\beta_i \neq 0$ for $1 \leq i \leq r$. Also, $r > 1$, or else the first equation would read $\beta_1 \sigma_1(\alpha_1) = 0$, whence $\beta_1 = 0$. As this is a homogenous system of equations, we may also assume $\beta_r = 1$. Thus the system of equations takes the form

$$\sum \sigma_j(\alpha_i) \beta_j \sigma_j(\alpha_r) = 0$$

As not all β_i are in F, we can assume without loss of generality that β_1 is not in F. As F is the fixed field of G, there is $\sigma_k \; inG$ for which $\sigma_k(\beta_1) \neq \beta_1$.

Applying σ_k to the system of equations yields

$$\sum_{i=1}^{r-1} \sigma_k(\sigma_j(\alpha_i)) \sigma(\beta_i) + \sigma_k(\sigma_j(\alpha_r)) = 0$$

As G is a group, applying this just permutes the indices of σ_i, so the system is given by

$$\sum_{i=1}^{r-1} \sigma_j(\alpha_i) \sigma_k(\beta_i) + \sigma_j(\alpha_r) = 0$$

Subtracting these two expressions yield

$$\sum_{i=1}^{r-1} \sigma_j(\alpha_i)\left[\beta_i - \sigma_k(\beta_i)\right] = 0$$

in which case

$$(\beta_1 - \sigma_k(\beta_1), \ldots, \beta_{r-1} - \sigma_k(\beta_{r-1}), 0, \cdots, 0)$$

is a non-trivial solution (as $\beta_1 \neq \sigma_k(\beta_1)$). However, this contradicts our initial assumptions that $(\beta_1, \ldots, \beta_r, 0, \ldots, 0)$ was a non-trivial solution with minimal number of non-zero terms.

Thus $(E : F) = n$. □

Corollary 6. *If $G \subset \mathrm{Aut}(E)$ is a finite group and $F = E^G$, then $G = \mathrm{Gal}(E/F)$.*

Proof. By the theorem we have $(E : F) = n = |G|$.

Suppose for contradiction there is $\sigma \in \mathrm{Aut}(E) \setminus G$ that fixes F. This implies F is fixed by $n+1$ distinct elements of $\mathrm{Aut}(E)$ and thus $(E : F) \geq n+1$, which is a contradiction. □

Corollary 7. *Let G_1, G_2 be finite subgroups of $\mathrm{Aut}(E)$ with $G_1 \neq G_2$. Then $E^{G_1} = F_1 \neq F_2 = E^{G_2}$.*

Proof. If $F_1 = F_2 = F$, then both G_1 and G_2 are equal to $\mathrm{Gal}(E/F)$. □

4.6 Separable Extensions

Definition 54.

1. Let $p(x) \in F[x]$. We say $p(x)$ is *separable* if its irreducible factors do not have repeated roots (in any splitting field).
2. If E/F is an extension, an element $\alpha \in E$ is *separable* (over F) if its is the root of a separable polynomial $f(x) \in F[x]$. This is the case exactly when its minimal polynomial $f_\alpha(x)$ is separable.
3. E/F is *separable* if every $\alpha \in E$ is separable over F.

Theorem 26. *An extension E/F is a (finite) Galois extension if and only in E is the splitting field of a separable polynomial $p(x) \in F[x]$.*

Proof. Suppose that E is the splitting field of $p(x) \in F[x]$ with p separable.

In the case that all the roots of $p(x)$ lie in F, then $E = F$ and so F is the fixed field of $G = \{\mathrm{id}\}$. We induct on the number n of roots of $p(x)$ not lying in F.

In the case $n > 1$ (note $n = 1$ means all roots of p are in F), suppose the theorem holds for all pairs of fields E/F and separable polynomials p with less than n roots outside of F. We factor $p(x)$ into irreducible factors in $F[x]$, so $p(x) = p_1(x) \cdots p_r(x)$. Without loss of generality we can take $\deg p_1(x) = s > 1$. If α_1 is a root of $p_1(x)$ lying outside of F, we have $E \supset F(\alpha_1) \supset F$ and $(F(\alpha_1) : F) = s$. In this case, $E/F(\alpha_1)$ remains the splitting field of $p(x)$ and the number of roots outside $F(\alpha_1) < n$. By the inductive hypothesis, we have $E/F(\alpha_1)$ Galois.

Consider $G_1 = \mathrm{Gal}(E/F(\alpha_1))$. As $F(\alpha_1) = E^{G_1}$, we have $\beta \in E \setminus F(\alpha_1)$ moved by some element of G_1. As $p_1(x)$ is separable, it has s distinct roots, so for every i there is $\sigma_i \in \mathrm{Gal}(E/F)$ with

$$\sigma_i(\alpha_1) = \alpha_i$$

and $\sigma_i : F(\alpha_1) \to F(\alpha_i)$ leaves F invariant. Note E remains splitting field of $p(x)$ over $F(\alpha_i)$. Thus, every σ_i extends to an element of $\mathrm{Aut}(E)$ leaving F fixed and sending α_1 to α_i.

Let $G = \mathrm{Gal}(E/F)$, and suppose $\theta \in E^G$. As $\mathrm{Gal}(E/F(\alpha_1)) \subset \mathrm{Gal}(E/F)$, and $E^{G_1} = F(\alpha_1)$, we have $\theta \in E^{G_1} = F(\alpha_1)$, so we can write

$$\theta = c_0 + c_1\alpha + \cdots + c_{s-1}\alpha_1^{s-1}$$

where $c_i \in F$. Applying σ_i to this we have

$$c_0 + c_1\alpha_1 + \cdots + c_{s-1}\alpha_1^{s-1} = \theta = \sigma_1(\theta) = c_0 + c_1\alpha_i + \cdots + c_{s-1}\alpha_i^{s-1}$$

for all i. This implies the polynomial

$$(c_0 - \theta) + c_1x + \cdots + c_{s-1}x^{s-1}$$

has α_i as roots, which is greater than the degree of the polynomial whence the polynomial is 0. But this implies $\theta = c_0$ for some $c_0 \in F$.

This gives us that $F = E^{\mathrm{Gal}(E/F)}$ and E/F is Galois.

For the other direction, assume E/F is finite Galois. We use the following lemma:

Lemma 6. *If E/F is finite Galois, then E/F is separable and normal.*

Proof. Let $G = \mathrm{Gal}(E/F) = \{\sigma_1, \ldots, \sigma_n\}$ with $\sigma_1 = \mathrm{id}$.

Let $\alpha \in E$ and let $\alpha_1, \ldots, \alpha_r$ be the distinct elements of the G-orbit of α with $\alpha_1 = \alpha$. This means that $f(x) = (x - \alpha_1)\cdots(x - \alpha_r)$ is fixed by G, so $f(x) \in F[x]$ as $F = E^G$.

We claim that $f(x)$ is irreducible. Indeed, if $g(x) \mid f(x)$, then $g(x)$ has some root α_i. But by applying the σ_i we see $g(x)$ has all α_i as roots, so $\deg g(x) \geq \deg f(x)$ and $f(x)$ is irreducible and $f(x)$ is the minimal polynomial of α and is separable and splits completely.

This implies E/F is separable and normal. □

Now suppose E/F is finite Galois with $w_1, \ldots, w_t$ a basis of E/F. Let $f_i(x) \in F[x]$ be the minimal polynomial of w_i. By the lemma this is separable. The product $f(x) = f_1(x)\cdots f_t(x) \in F[x]$ is separable and E is the splitting field of $f(x)$ as each f_i splits completely in E by normality. □

Looking at the logic of the proof we have

$$E/F \text{ is the splitting field of a sep. poly } \Rightarrow E/F \text{ is finite Galois } \Rightarrow$$

$$E/F \text{ is finite, separable, and normal } \Rightarrow E/F \text{ is the splitting field of a sep. poly}$$

Because of this interconnection between splitting fields of separable polynomials and Galois extensions, we make the following definition:

Definition 55. If E/F is the splitting field of a (separable) polynomial $f(x) \in F[x]$ we will call $\mathrm{Gal}(E/F) = G_f$ the Gaolis group of the polynomial $f(x)$.

We now arrive at the Fundamental Theorem of finite Galois Theory

Theorem 27 (Fundamental Theorem of Finite Galois Theory)**.** *Let $p(x) \in F[x]$ be separable, E/F the splitting field of $p(x)$, and $G = \mathrm{Gal}(E/F) = G_p$. Then:*

1. *Every intermediate field $E \supset K \supset F$ is the fixed field $E^H = K$ for a unique subgroup $H < G$. Thus $K = E^H$ and $H = \mathrm{Gal}(E/K)$.*

2. *K/F is Galois if and only if $H \trianglerighteq G$. In this case $\mathrm{Gal}(K/F) \cong G/H$.*

3. If $E \supset K \supset F$ and $H = \mathrm{Gal}(E/K)$ then $(K : F) = (G : H)$ and $(E : K) = |H|$.

Proof. (1): If $E \supset K \supset F$, then E is still the splitting field of $p(x) \in F[x]$ over K. Thus E/K is Galois; if $H = \mathrm{Gal}(E/K)$ then $H < G$ and $K = E^H$. As distinct subgroups of G has distinct fixed fields we have a bijection between intermediate fields and subgroups of H by

$$K \mapsto \mathrm{Gal}(E/K) = H$$

and

$$H \mapsto E^H$$

(3): Using the correspondence established above, we have

$$(E : K) = |\mathrm{Gal}(E/K)| = |H|$$

and also

$$(K : F) = \frac{(E : F)}{(E : K)} = \frac{|G|}{|H|} = |G/H| = (G : H)$$

(2): Consider an intermediate field $E \supset K \supset F$ and $H = \mathrm{Gal}(E/K)$.

In the case $\sigma \in \mathrm{Gal}(E/F) = G$ then $\sigma(E) = E$ and $\sigma(K) \subset \sigma(E) = E$ is another intermediate field. This gives us the following diagram:

$$\begin{tikzcd}
E \arrow[r, "\sigma"] \arrow[d] & E \arrow[d] \\
K \arrow[r, "\sigma"] & \sigma(K) \\
& F \arrow[ul] \arrow[ur]
\end{tikzcd}$$

If $\tau \in H$ then $\tau(K) = K$ and $\sigma\tau(K) = \sigma(K)$. Thus, elements of the same coset of G/H give the same intermediate field $\sigma(K)$.

We also have a partial converse: If σ_1 and σ_2 lie in distinct cosets of G/H then σ_1 and σ_2 must give distinct isomorphism $K \to \sigma_i(K)$. If this were not the case $\sigma_1\sigma_2^{-1}$ would fix K and therefore be in H.

If K_1 and K_2 are two intermediate fields and $\sigma : K_1 \to K_2$ fixing F, then E is still the splitting field of $p(x)$ over F, and we can therefore extend σ to an element of $\mathrm{Gal}(E/F)$. This gives the diagram

$$\begin{tikzcd}
E \arrow[r, "\sigma"] \arrow[d] & E \arrow[d] \\
K_1 \arrow[r, "\sigma"] & K_1 \\
& F \arrow[ul] \arrow[ur]
\end{tikzcd}$$

All of this implies that the distinct embeddings $\sigma : K \hookrightarrow E$ all come restrictions of $\sigma \in \mathrm{Gal}(E/F)$. As these embeddings are invariant under H and distinct cosets of G/H give different embeddings, we then have the number of such embeddings as $|G/H|$.

Conjugation by elements of $\mathrm{Gal}(E/F)$ acts nicely on $H = \mathrm{Gal}(E/K)$, as for $\sigma \in \mathrm{Gal}(E/F) = G$ we have

$$\mathrm{Gal}(E/\sigma(K)) = \sigma H \sigma^{-1} = \sigma\, \mathrm{Gal}(E/K)\sigma^{-1}$$

as elements of $\sigma(K)$ are fixed under this automorphisms and equality follows by the counting argument

$$|\mathrm{Gal}(E/\sigma(K))| = (E : \sigma(K)) = (E : K) = |H| = \left|\sigma H \sigma^{-1}\right|$$

Now suppose that K/F is Galois. Then we have

$$|\mathrm{Gal}(K/F)| = (K : F)$$

The converse is also true. If $F' = K^{\mathrm{Gal}(K/F)}$ then $K \supset F' \supset F$. As $(K : F') = |\mathrm{Gal}(K/F)|$ and $|\mathrm{Gal}(K/F)| = (K : F)$, we have $(K : F') = (K : F)$ and $F' = F$. Thus K is Galois over F if and only if $(K : F) = |\mathrm{Gal}(K/F)|$.

We have seen by part 3 that

$$|\mathrm{Gal}(K/F)| = (K : F) = (G : H) = |G/H|$$

and this last number is the number of distinct embeddings $\sigma : K \hookrightarrow E$. As the number of distinct embeddings $\sigma : K \hookrightarrow E$ is the number of isomorphism $\sigma : K \to K$ over F, we must have each $\sigma(K) = K$ for $\sigma \in \mathrm{Gal}(E/F)$.

Therefore we have K/F is Galois if and only if $|\mathrm{Gal}(K/F)| = (K : F)$ if and only if $\sigma(K) = K$ ofr all $\sigma \in \mathrm{Gal}(E/F)$ if and only if $\sigma H \sigma^{-1} = H$ for all $\sigma \in \mathrm{Gal}(E/F)$ if and only if $H \trianglelefteq G$.

Finally, as the distinct embeddings $K \hookrightarrow E$ over F is isomorphic to G/H and the distinct automorphism of K fixing F (which is $\mathrm{Gal}(K/F)$, we have $\mathrm{Gal}(K/F) \cong G/H$. □

5 Finite Fields

Theorem 28. *Let F be a field, S a finite subgroup of $F^\times$. Then S is a cyclic group.*

Proof. By the classification of finite abelian groups, we have $S \cong C_1 \oplus \cdots C_t$ where C_i is the cyclic group of order i, and $i \mid i+1$. Note then that any element of S satisfies $x^t - 1 = 0$. Thus there are at most t elements of S, as this has at most t distinct roots. This implies by counting that $S = C_t$. □

Corollary 8. *If F is a finite field, $F^\times$ is cyclic.*

5.1 The Characteristic of a Field

Let F be an arbitrary field. We always have a map $\epsilon : \mathbb{Z} \to F$ by $n \mapsto n \cdot 1$. This has a kernel, and as $\mathbb{Z}$ is a PID, it is generated by some prime p or 0. In the case the kernel is trivial, we say F has characteristic 0. In the case the kernel is generated by p, we say $\mathrm{Char}\, F = p > 0$.

Note that if $\mathrm{Char}\, F = p$, then for any $x \in F$, $px = \sum_{i=1}^{p} x = 0$.

Let F be a finite field.

Theorem 29. *If F is a finite field, then* $\mathrm{Char}\, F = p$ *for some prime p and $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p \subset F$.*

Proof. The kernel of the map $\epsilon : \mathbb{Z} \to F$ must be non-trivial by a counting argument, so $\mathrm{Char}\, F > 0$. This map then induces an embedding $\mathbb{F}_p \subset F$. □

Corollary 9. *IfF is a finite field then $|F| = p^v = q$ for some $v \geq 1$.*

Proof. Note that F is a finite dimensional vector space over $\mathbb{F}_p$ by the above theorem. From linear algebra we then have $|F| = p^v$ for some v. □

We also have that finite fields are splitting fields of a canonical polynomial over $\mathbb{F}_p$.

Theorem 30. *Let F be a finite field of characteristic p. Then F is the splitting field of $x^q - x$ over $\mathbb{F}_p$.*

Proof. We have that $F^\times$ is cyclic of order $q-1$, so every $\alpha \in F^\times$ is a root of $x^{q-1} - 1 = 0$ and by factorization we have

$$x^{q-1} - 1 = \prod_{\alpha \in F^\times} (x - \alpha)$$

We then have $x^q - x = \prod_{\alpha \in F}(x - \alpha)$. Therefore we see F is the splitting field of $x^q - x$ over $\mathbb{F}_p$. □

Corollary 10. *If $|F| = q = p^v$, then $F = \{\alpha : \alpha \text{ is a root of } x^q - x\}$.*

Corollary 11. *Any two finite fields of the same order are isomorphic.*

Proof. We've seen both of these fields are the splitting fields of the polynomial $x^q - x$ over their prime fields. As the prime fields are isomorphic by $1 \mapsto 1$, the uniqueness of splitting fields gives these two fields are isomorphic. □

5.2 Separability and Differentiation

Let F be an arbitrary field. If $f(x) = a_0 + a_1 x + \cdots + a_n x^n \in F[x]$, the we can define a formal derivative by $f'(x) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1}$.

This formal derivative satisfies many of the same properties of analytic differentiation:

Proposition 17.

1. $(f_1 + f_2)' = f_1' + f_2'$
2. $(f_1 \cdot f_2)' = f_1 \cdot f_2' + f_1' \cdot f_2$
3. $(f^n)' = n f^{n-1} \cdot f'$

This formal derivative is an important tools for determining the separability of irreducible polynomials.

Theorem 31. *Let $f \in F[x]$. The following are equivalent:*

1. *$f(x)$ has repeated roots.*
2. *In a splitting field E of f, $f(x)$ and $f'(x)$ have a common root.*
3. *$\gcd(f, f') > 1$ in $F[x]$.*

Proof. (1) $\Leftrightarrow$ (2): Let α be a root of $f(x) = 0$ in E. Suppose the multiplicity of α is $k \in \mathbb{N}$. Thus

$$f(x) = (x - \alpha)^k q(k)$$

in $E[x]$ with $q(alpha) \neq 0$. By the properties of the derivative

$$f'(x) = (x - \alpha)^k q'(x) + k(x - \alpha)^{k-1} q(x)$$

We then see α is a root if and only if $k > 1$. Thus f has repeated roots if and only if f and f' have a common root in a splitting field.

(2) $\Leftrightarrow$ (3): If α is a root of both f and f', then the minimal polynomial f_α divides both f and f', so $\gcd(f, f') > 1$.

On the other hand, if $\gcd(f, f') > 1$, then as f splits, so does the common divisor of f, f', and this has a root, whence f, f' have a common root. □

Corollary 12. *Let F be a field of characteristic 0. Then every irreducible polynomial $f(x) \in F[x]$ is separable.*

Proof. Let $f(x) \in F[x]$ be irreducible. Without loss of generality, f is monic. Note then that $f'(x) \neq 0$ and has degree one less than that of f. As f is irreducible, it is the minimal polynomial of all of its roots. By the minimality of this polynomial, any root of f in E a splitting field cannot be a root of $f'(x)$. Thus f is separable. □

N.B. In the case $\operatorname{Char} F = p > 0$ this can fail, specifically in the case $f' = 0$. This happens exactly if f is of the form $g(x^p)$ for some polynomial g.

Definition 56. A field F is *perfect* if every irreducible polynomial $f \in F[x]$ is separable.

Corollary 13. *Any field of characteristic 0 is perfect.*

In the homework we will show the following:

Theorem 32. *Any finite field is perfect.*

Thus any non-perfect field must be an infinite field of positive characteristic.

5.3 Cyclotomic Fields

A *cyclotomic field* is a field generated by appending a root of unity to a base field F. We will see that these extensions have abelian Galois groups. In fact over $\mathbb{Q}$ any Abelian extension (an extension with abelian Galois groups) is the subfield of a cyclotomic field.

For F a field, consider the polynomial $x^n - 1 \in F[x]$. In the case n is not divisible by the characteristic of the field, this separable as its derivative is nx^{n-1} which only has 0 as a root, and this is not the root of $x^n - 1$. For the remainder of the section we will thus assume n is not divisible by the characteristic of the field.

Let E/F be the splitting field of $x^n - 1$. Then E is called the nth *cyclotomic field over* F. This is a Galois extension as the splitting field of a separable polynomial. The set of roots form a finite multiplicative subgroup of $E^\times$ and therefore is cyclic. Thus $E = F(\zeta_n)$ where ζ_n is any generator of this subgroup.

Theorem 33. *The Galois group* $\operatorname{Gal}(E/F)$ *is Abelian. If $n = q$ is prime (and not* $\operatorname{Char} F$*) then* $\operatorname{Gal}(E/F)$ *is cyclic.*

Proof. For $\sigma \in \operatorname{Gal}(E/F)$ we have $\sigma(\zeta_n)$ is another root of $x^n - 1 = 0$. There is thus a natural number $1 \leq n_\sigma < n$ with $\sigma(\zeta_n) = \zeta_n^{n_\sigma}$ (note that $n_\sigma \neq 0$, or else the kernel σ would be non-trivial and thus all of K). We also see that the map $\sigma \mapsto n_\sigma$ is a homomorphism $\operatorname{Gal}(E/F) \to (\mathbb{Z}/n\mathbb{Z})^\times$. This is injective, as σ is defined by its action on ζ_n; therefore $\operatorname{Gal}(E/F)$ is isomorphic to a subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$ and thus abelian.

In the case $n = q$ prime $(\mathbb{Z}/q\mathbb{Z})^\times$ is cyclic and thus so is any subgroup. □

5.4 Noether's Equations

The following material was proven by Emmy Noether. The modern formulation of these results are now given in group or Galois cohomology, although we will not cover that here.

Suppose E is a field and G a finite subgroup of $\operatorname{Aut}(E)$.

Let $\sigma \mapsto a_\sigma$ be any map $G \to E^\times$. The set of elements $\{a_\sigma : \sigma \in G\}$ is called a *solution to Noether's equations* if

$$a_\sigma \cdot \sigma(a_\tau) = a_{\sigma\tau}$$

for all $\sigma, \tau \in G$.

Theorem 34 (Speiser)**.** *The set $\{a_\sigma : \sigma \in G\}$ is a solution to Noether's equations if and only if there exists $\beta \in E^\times$ such that*

$$a_\sigma = \frac{\beta}{\sigma(\beta)}$$

for all $\sigma \in G$.

Proof. Note that the automorphisms $\tau \in G$ are linearly independent over E by Dedekind's independence of characters. As $a_\tau \neq 0$ we thus have some $\gamma \in E^\times$ with

$$\sum_{\tau \in G} a_\tau \tau(\gamma) = \beta \neq 0$$

Applying σ to β yields

$$\sigma(\beta) = \sum_{\tau \in G} \sigma(a_\tau)\sigma\tau(\gamma)$$

and multiplying by a_σ gives

$$a_\sigma \sigma(\beta) = \sum_{\tau \in G} a_\sigma \sigma(a_\tau)\sigma\tau(\gamma)$$

In the case Noether's equations hold we get

$$a_\sigma \sigma(\beta) = \sum_{\tau \in G} a_{\sigma\tau}\sigma\tau(\gamma) = \sum_{\tau \in G} a_\tau \tau(\gamma) = \beta$$

Thus

$$a_\sigma = \frac{\beta}{\sigma(\beta)}$$

Now in the case $a_\tau = \frac{\beta}{\sigma(\beta)}$ we then have

$$a_\sigma \cdot \sigma(a_\tau) = \frac{\beta}{\sigma(\beta)} \cdot \frac{\sigma(\beta)}{\sigma\tau(\beta)} = \frac{\beta}{\sigma\tau(\beta)} = a_{\sigma\tau}$$

and thus Noether's equations hold. □

N.B. In terms of group cohomology, this theorem says that $H^1(G, E^\times) = 1$.

Suppose that each a_σ lies in the fixed field $F = E^G$. In this case, we see the map $\chi : G \to F^\times$ given by $\chi(\sigma) = a_\sigma$ with a_σ satisfying Noether's equations is a character of G with values in $F^\times$ because

$$\chi(\sigma\tau) = a_{\sigma\tau} = a_\sigma \sigma(a_\tau) = a_\sigma a_\tau = \chi(\sigma)\chi(\tau)$$

By reversing this argurment, and character $\chi : G \to F^\times$ defines a solution to Noether's equations.

This gives us the following theorem.

Theorem 35. *Let E/F be a Galois extension with $G = \mathrm{Gal}(E/F)$. Then for each character $\chi : G \to F^\times$ there is $\beta \in E^\times$ such that*

$$\chi(\sigma) = \frac{\beta}{\sigma(\beta)}$$

and conversely, if $\beta/\sigma(\beta) \in F$ for all $\sigma \in G$ the $\chi_\beta(\sigma) = \beta/\sigma(\beta)$ is a character of G. Moreover, if r is the least common multiple of the orders of $\sigma \in G$, then $\beta^r \in F$.

Proof. We have seen above that the first part of the theorem holds. Now suppose r is the least common multiples of order of $\sigma \in G$. To see that $\beta^r \in F$ we will show $\sigma(\beta^r) = \beta^r$ for all $\sigma \in G$. Indeed

$$\frac{\beta^r}{\sigma(\beta^r)} = \left(\frac{\beta}{\sigma(\beta)}\right)^r = \chi(\sigma)^r = \chi(\sigma^r) = \chi(e) = 1$$

□

5.5 Kummer Theory

In the case our base field has enough roots of unity, we have a characterization of Abelian extensions.

Let F be a field containing n distinct n^{th} roots of 1. Thus we must have the characteristic of F not dividing n. Let $\mu_n(F)$ be the (multiplicative) group of distinct n^{th} roots of 1 in F.

Definition 57. A *Kummer extension* E of F is the splitting field of a polynomial

$$f(x) = (x^n - a_1) \cdots (x^n - a_r)$$

with $a_1, \ldots, a_r \in F$. This gives $E = F(\sqrt[n]{a_1}, \ldots, \sqrt[n]{x^n - a_r})$.

In the case that $a_i \neq a_j$ for $i \neq j$ we have f separable, for if $\alpha_i^n = a_i$ then for any $\epsilon_k \in \mu_n(F)$ we have $(\epsilon_k \alpha_i)^n = a_i$, so there are exactly n roots of $x^n - a_i$ over F, as $|\mu_n(F)| = n$. As E is a splitting field of a separable polynomial, we then have E/F is a Galois extension.

The action of $\mathrm{Gal}(E/F)$ is determined by its action on α_i. For $\sigma \in \mathrm{Gal}(E/F)$ we thus have $\sigma(\alpha_i) = \epsilon_{\sigma,i}\alpha_i$, as $\sigma(\alpha_i)$ is anothr root fo $x^n - a_i$. Then for $\sigma, \tau \in \mathrm{Gal}(E/F)$ we have

$$\sigma(\tau(\alpha_i)) = \sigma(\epsilon_{i,\tau}\alpha_i) = \epsilon_{i,\tau}\epsilon_{i,\sigma}\alpha_i = \tau(\sigma(\alpha_i))$$

as $\mathrm{Gal}(E/F)$ is an Abelian group.

Theorem 36. *If F contains n distinct n^{th} roots of 1 and E is a Kummer extension of F then*

1. *E/F is a Galois extension.*
2. *E/F is Abelian.*
3. *$\mathrm{Gal}(E/F)$ has exponent m dividing n.*

Proof. We have left to prove 3. Indeed, note that $\sigma^k(\alpha_i) = \epsilon_{i,\sigma}^k \alpha_i$. As each ϵ_i has order dividing m as $|\mu_n(F)| = m$, we then have the order of σ dividing n. □

Corollary 14. *Let q be a prime and F contains q distinct q^{th} roots of 1. Let E be the splitting field of $x^q - a$ with $a \in F$. Then either $E = F$ or $x^q - a$ is irreducible and $\mathrm{Gal}(E/F)$ is cyclic of order q.*

Proof. From part 3 of the above theorem, we have $\mathrm{Gal}(E/F)$ has exponent dividing q so is either 1 or q. In the first case $\mathrm{Gal}(E/F)$ is trivial, so $E = F$. In the second, $(E : F) = q$, so $x^q - a$ is irreducible. □

Interestingly, the three conditions in the above theorem actually determine Kummer extensions. Suppose that E/F is an Abelian, Galois extension and if m is the exponent of $\mathrm{Gal}(E/F)$ then F contains m distinct m^{th} roots of 1.

Let $\mu_m(F)$ be the mth roots of unity in F and $\mathrm{Gal}(E/F)$. Let

$$\widehat{G} = \left\{\chi : G \to \mu_m(F) \subset F^\times\right\}$$

Note that as G has expoenent m, these are actually all the characters of G taking values in F.

Proposition 18. *Since $m = \exp(G)$, $\widehat{G} \cong G$. Moreover, for any $\sigma \in G$, $\sigma \neq e$ there is $\chi \in \widehat{G}$ with $\chi(\sigma) \neq 1$.*

Proof. By the structure theorem of finite Abelian groups we have

$$G \cong C_{m_1} \times \cdots \times C_{m_t}$$

where $m_i \mid m_{i+1}$ and $m_t = m$. Let σ_i be a generator of of C_{m_i}. Then every element of G can be written uniquely in the form

$$\sigma = \sigma_1^{\nu_1} \cdots \sigma_t^{\nu_t}$$

where ν_i is taken modulo m_i.

Define $\chi_i : G \to \mu_m(F)$ by

$$\chi_i(\sigma_j) = \begin{cases} \epsilon_i & j = i \\ 1 & j \neq i \end{cases}$$

where ϵ_i is some primitive m_i^{th} root of unity. For $\chi \in \widehat{G}$ we can then write $\chi(\sigma_i) = \epsilon_i^{\mu_i} = \chi_i(\sigma_i)^{\mu_i}$. Thus we have that any character of G takes the form

$$\chi = \chi_1^{\mu_1} \cdots \chi_t^{\mu_t}$$

and any such representation defines a character of G. As χ_i has order m_i we then have $\widehat{G} \cong C_{m_1} \times \cdots \times C_{m_t} \cong G$.

For $\sigma \in G$ not the identity, we can write $\sigma = \sigma_1^{\nu_i} \cdots \sigma_t^{\nu_t}$, and one of these $\nu_i \neq 1$. Thus $\chi_i(\sigma) = \epsilon_i^{\nu_i} \neq 1$. □

Define

$$A = \left\{\alpha \in E^\times : \alpha^m \in F\right\}$$

Then A is a multiplicative subgroup of $E^\times$ and $F^\times \subset A$. Let

$$A^m = \{\alpha^m : \alpha \in A\} \subset F^\times \quad \text{and} \quad F^{\times,m} = \left\{\alpha^m : \alpha \in F^\times\right\} \subset A^m$$

The following proposition gives a convenient way of calculating $G = \mathrm{Gal}(E/F)$.

Proposition 19. *1. $A/F^\times \cong A^m/F^{\times,m}$*

2. $A/F^\times \cong \widehat{G}$

Proof. 1. Consider the map $A \to A^m$ given by taking m^{th} powers (this is surjective), and compose with the quotient $A^m \to A^m/F^{\times,m}$. Let K be the kernel of this map. We get the following exact sequence:

$$1 \to K \to A \to A^m/F^{\times,m} \to 1$$

In the case $\beta \in K$ there is $b^m \in F^{\times,m}$ with $\beta^m = b^m$. This implies β is a root of

$$x^m - b^m = 0$$

the roots of $x^m - b^m = 0$ are given by $\epsilon_i b$ where ϵ_i are the mth roots of unity in F. Thus we have $\beta = \epsilon_i b$ for some ϵ_i. In particular, $\beta \in F^\times$ and $K \subset F^\times$. As $F^\times \subset K$ we then have $K = F^\times$ so

$$A/F^\times \cong A^m/A^m/F^{\times,m}$$

2. Fix $\alpha \in A$. For each $\sigma \in G = \mathrm{Gal}(E/F)$ we have

$$\left(\frac{\alpha}{\sigma(\alpha)}\right)^m = \frac{\alpha^m}{\sigma(\alpha^m)} = \frac{\alpha^m}{\alpha^m} = 1$$

as $a^m \in F$. Thus we see that $\alpha/\sigma(\alpha) \in \mu_m(F)$ and

$$\chi_\alpha(\sigma) = \frac{\alpha}{\sigma(\alpha)}$$

is a character of G with values in F [this is from the theorem following Speiser's Theorem]. The map $\alpha \mapsto \chi_\alpha$ is a homomorphism $A \to \widehat{G}$ since

$$\chi_\alpha\chi_\beta(\sigma) = \chi_\alpha(\sigma)\chi_\beta(\sigma) = \frac{\alpha}{\sigma(\alpha)}\frac{\beta}{\sigma(\beta)} = \frac{\alpha\beta}{\sigma(\alpha\beta)} = \chi_{\alpha\beta}(\sigma)$$

Moreover, the same theorem states that every such character arises in this way.

We have $(\alpha/\sigma(\alpha))^m = 1$ so $\alpha^m = \sigma(\alpha^m)$ for all $\sigma \in G$. This implies $\alpha^m \in F$ and so $\alpha \in A$. Thus we have

$$A \to \widehat{G} \to 1$$

by $\alpha \mapsto \chi_\alpha$.

The kernel of this map occurs exactly when $\frac{\alpha}{\sigma(\alpha)} = 1$ for all σ, and this is the case exactly when $\alpha \in F^\times$. Thus we have $A/F^\times \cong \widehat{G}$.

□

By this proposition we have that $A/F^\times$ and $A^m/F^\times$ are finite groups and we have isomorphisms

$$A^m/F^{\times,m} \cong A/F^\times \cong \widehat{G} \cong G$$

The left hand side is internal to F and the right hand side is $\mathrm{Gal}(E/F)$. Thus, we have some way of characterizing $\mathrm{Gal}(E/F)$ in terms of the base field F itself.

We can now finish our characterization of Kummer extensions.

Theorem 37. *Let E/F be a finite algebraic extension. Then E/F is a Kummer extension of F if and only if*

1. *E/F is a fintie Galois extension,*
2. $\mathrm{Gal}(E/F)$ *is abelian, and*
3. *if m is the exponent of* $\mathrm{Gal}(E/F)$ *then F contains m distinct m^{th} roots of unity.*

Proof. We have seen the forward implication already.

Suppose all three conditions hold. Let A be defined as above. As $A/F^\times$ is a finite group we can write

$$A = \alpha_1 F^\times \cup \alpha_2 F^\times \cup \cdots \cup \alpha_t F^\times$$

by choosing representatives of cosets of $A/F^\times$. Then $t = |A/F^\times| = |G| = (E : F)$.

As $\alpha_i \in A$ we have α_i the root of the polynomial

$$x^m - \alpha_i^m \in F[x]$$

and this splits completely in E by the third assumption.

Moreover, we claim that E is the splitting field of

$$\prod_{i=1}^{t}(x^m - \alpha_i^m) \in F[x]$$

If this is not the case, then $F(\alpha_1, \ldots, \alpha_t)$ is an intermediate field strictly between F and E. Thus there is some $\sigma \neq e$ that is the identity on $F(\alpha_1, \ldots, \alpha_t)$. In this case, there is a character $\chi \in \widehat{G}$ with $\chi(\sigma) \neq 1$. This character take the form $\chi(\sigma) = \frac{\alpha}{\sigma(\alpha)} \neq 1$ for some $\alpha \in A$. This gives that $\sigma(\alpha) \neq \alpha$, so $\alpha \notin F(\alpha_1, \ldots, \alpha_t)$. However, $A = \alpha_1 F^\times \cup \cdots \cup \alpha_t F^\times$ so $A \subset F(\alpha_1, \ldots, \alpha_t)$, which is a contradiction.

Therefore $E = F(\alpha_1, \ldots, \alpha_n)$ and is thus a Kummer extension of F. □

In the case that m is a prime, the following corollary is immediate.

Corollary 15. *If E/F is Galois with $(E : F) = q$ prime and F contains distinct q^{th} roots of unity, then E is the splitting field of an irreducible polynomial $x^q - \alpha \in F[x]$.*

N.B. If fact, assuming that F contains m distinct m^{th} roots of unity, there is an order reversing bijection between subgroups N where $F^\times \supset N \supset F^{\times,m}$ and abelian extensions E/F with exponent dividing m.

5.6 Simple Extensions

Definition 58. An extension E/F is called *simple* if E is generated over F be a single element. This element is called a *primitive* element for E/F.

The following theorem characterizes when a finite extension is a simple.

Theorem 38 (Steinitz)**.** *Let E/F be a finite extesnion. Then E/F has a primitive element if and only if there are only a finite number of intermediate fields between F and E.*

Proof. Suppose that $E = F(\alpha)$. Let K be an intermediate field. Let $f(x)$ be the minimal polynomial of α over F and $g(x)$ the minimal polynomial of α over K. By definition of the minimal polynomial, we have $g(x)$ divides $f(x)$ in $K[x]$ and thus $E[x]$.

Let K' be the subfield of E generated over F by coefficients of $g(x)$. Since $g(x) \in K[x]$ we have $K' \subset K$ and $g(x)$ is also the minimal polynomial of α over K'. Thus $E = F(\alpha) = K(\alpha) = K'(\alpha)$ and $(E : K) = \deg(g) = (E : K')$. Thus $K = K'$.

Therefore the intermediate fields of E and F are generated by the coefficients of the monic factors $g(x)$ of $f(x)$ in $E[x]$. As there only a finitely many such factors there can only by a finite number of intermediate fields.

For the other direction, suppose there are only finitely many intermediate fields.

In the case F (and thus E) are finite fields, $E^\times$ is a cyclic group and thus a generator of the cyclic group is necessarily a primitive element.

Thus assume that E and F are infinite fields. Let $E = F(\alpha_1, \ldots, \alpha_n)$. We induct on n. The case $n = 1$ gives the primitve element.

Now suppose $n = 2$, so $E = F(\alpha, \beta)$. There are intermediate fields of the form

$$K_t = F(\alpha + t\beta)$$

for all $t \in F$. By hypothesis there are only finitely many of these intermediate fields. There is therefore a pair $t_1 \neq t_2 \in F$ for which $F(\alpha + t_1\beta) = F(\alpha + t_2\beta)$.

Note that

$$\beta = \frac{(\alpha + t_1\beta) - (\alpha + t_2\beta)}{t_1 - t_2} \in F(\alpha + t_1\beta)$$

and thus $\alpha = (\alpha + t_1\beta) - t_1\beta \in F(\alpha + t_1\beta)$. Therefore we have $E = F(\alpha + t_1\beta)$ and E is simple.

Inductively if every finite extension with $n-1$ generators is simple, and $E = F(\alpha_1, \ldots, \alpha_n)$, then $E = F(\alpha_1, \ldots, \alpha_{n-1})(\alpha_n) = F(\beta)(\alpha_n) = F(\gamma)$ for some γ. □

Corollary 16. *If E/F is a finite separable extension, E is simple. In particular finite Galois extensions are simple.*

Proof. Let $\omega_1, \ldots, \omega_n$ be a basis of E/F. Let $f_i(x) \in F[x]$ be the minimal polynomial of ω_i over F. Then

$$f(x) = f_1(x) \cdots f_n(x)$$

is separable. Let K be the splitting field of $f(x)$ over F. Then $K \supset E \supset F$ and K/F is Galois.

By the fundamental theorem of finite Galois theory, the number of intermediate fields of K/F is in bijection with subgroups of $\mathrm{Gal}(K/F)$ which is finite. Thus there are only finitely many intermeditate fields between K and F and thus finitely many between E and F. □